Adaptivity versus Global Query Orders under Complete Equivalence Constraints

DRAFT – author details pending

Affiliation and contact information pending.

Abstract

For deterministic zero-error classification, we compare adaptive equality queries with a fixed global query order that permits history-dependent permanent skips, under identical information. Each input class contains all equivalence partitions satisfying its stated equalities and inequalities; unconstrained blocks may merge. A computer-assisted classification proves equal optimal worst purchased-set cost for every normalized monotone cost on menus of at most six queries. The proof combines complete source and orbit enumeration, adaptive frontiers, priority-policy certificates, and explicit analytic reductions. A complete seven-query source has a strict gap at positive additive prices, with a direct order-obstruction proof. On that fixed source the maximal ratio is **5/4** for nonnegative additive costs, with a unique attaining price ray, and **4/3** for monotone submodular costs, attained by a coverage cost. Strictly increasing costs in the respective additive and submodular classes retain these suprema without attainment. Three submodular inequalities prove the latter upper bound. Unrestricted strictly monotone costs give an unbounded ratio on the same source. A specified complete eleven-query extension and its identifiable coordinate projections retain the additive upper bound **5/4**. For zero-error randomized policies, a separate complete-source boundary is four queries: at most three, every normalized monotone cost equals the maximum cheapest promise-certificate cost, whereas four positive-priced queries permit randomization benefit and a common-completion gap. We provide complete frontiers and exact, finitely checkable certificate obligations. These are query-model results; practical academic-evaluation gains are not established.

Keywords: query complexity, adaptivity, ordered decision diagrams, equivalence constraints, submodular costs

Internal preparation draft. Human authorship details, author review and final author declarations remain pending. This is not a submitted or accepted article. Use of the publisher's recommended template does not establish journal-specific formatting or anonymity compliance.

1 Introduction

An answer can change which query is worth buying next. We compare adaptive acquisition with one global order allowing causal permanent skips, with the same promise and costs. Each promise retains every equivalence partition consistent with its equalities and inequalities. Completeness has a concrete effect: general Horn supports already separate at four queries, while complete equivalence supports require seven.

Table 1 separates the computer-assisted universal boundary from fixed-source ratios, proved algebraically from complete frontiers. Equivalence is an abstract relation; statistical independence and historical scientific validity are not assumed.

Source quantifier	Cost class	Conclusion
Every complete source, $m \leq 6$	Normalized monotone	$\mathsf{A}_F = \mathsf{P}_F$
One complete source, $m = 7$	Strictly positive additive	$\mathsf{P}_F > \mathsf{A}_F$ is possible
That same fixed source	Nonnegative additive	Maximum ratio 5/4
That same fixed source	Monotone submodular	Maximum ratio 4/3
That same fixed source	Strictly monotone	Ratios unbounded
One specified source, $m = 11$, and its identifiable projections	Nonnegative additive	Upper 5/4

Table 1 All set costs are normalized and nonnegative; ratios require $\mathsf{A}_F > 0$. On the seven-query source, strictly positive additive and strictly increasing submodular costs retain the respective suprema without attainment. The eleven-query row concerns one specified source; Appendix F gives sixteen matching menus. No all-seven-source or all-eleven-source upper bound is asserted.

Related work.

Charikar et al. [1] compare deterministic priced queries with an input-specific cheapest certificate. Such guarantees can bound minimax cost, but require a strategy in the allowed class and a suitable cost comparison. Section 2.5 makes this connection explicit; the two objectives cannot be separated merely by their denominators. Section 3.4 gives a concrete fixed-source formula-cost comparison.

Positive-rational additive instances fit discrete function evaluation [2], whose logarithmic approximation concerns adaptive trees. Ordered diagrams permit omitted variables; maximum-path objectives [3, 4], compatible completions [5, 6], and average-path cofactoring [7] are established. Greiner et al. [8] separate expected-cost AND/OR strategies under fixed permutations with causal skips. Section 2.4 identifies our comparator.

Graphic-flat and multicut representations [9, 10] are standard. Transferring cut-edge cost results or inconsistent 2-CNF clause-search bounds [11] requires preserving our visible menu, purchased-set costs and classification objective; complete fibres need not be 2-CNF.

Partition learning [12, 13] reconstructs the whole partition under different class-count and round assumptions. Matroidal query bounds [14] concern full-cube independence recognition. Neither task is our ordered-skip classification problem.

For independent box values, Berger et al. [15] give an optimal fixed exploration order for expected maximum reward minus submodular set cost. Our worst-cost zero-error relational classification needs a separate argument; no reduction is asserted.

Completion complexity, defined by minimizing a measure over compatible total functions, is established context. Huffstutler et al. [16] distinguish promise-valid from domain-certifying certificates and record preservation of deterministic unit-query complexity under an optimal completion. Our purchased-set completion identities are supporting tree transformations, not a new completion framework.

A separate randomized extension in Appendix I proves a sharp four-query boundary for randomization benefit and for a randomized promise/common-completion gap. The zero-error objective is worst-input expected purchased-set cost, and a global order may be sampled with the private seed. This differs from the main deterministic adaptive/order comparison. Consistent-output and completion phenomena already occur in search and communication complexity [17–19]; the latter work also treats zero-sided error. We claim no new general completion framework or generic zero-error separation.

Candidate contributions are the exact complete-source boundaries and fixed-source cost bounds. Generic representations, dominance and certificate checking are supporting tools. Exact literature priority remains under review.

2 Model and complete policy recursion

2.1 Complete partition fibres

Let R be a finite record set. For each label $y \in \{A, B\}$, let C_y be a conjunction of $\mathsf{SAME}(r, s)$ and $\mathsf{DIFFERENT}(r, s)$ constraints, and let Ω_y contain *all* equivalence partitions satisfying C_y. Both fibres are nonempty. The number of equivalence classes is unrestricted.

A physical menu $Q = \{q_0, \ldots, q_{m-1}\}$ assigns each coordinate a pair of records. Its answer is 1 exactly when that pair is equivalent. Write $W_y \subseteq \{0,1\}^m$ for the projection of Ω_y onto the menu. Assume $W_A \cap W_B = \varnothing$. Distinct partitions can have the same answer vector. Distinct physical coordinates keep their separate cost identities even when their answers are equivalent.

Unlisted equalities do not imply inequalities: unconstrained blocks may merge. Arbitrary pruning of W_y, bounding the number of classes, or imposing a distribution changes the model.

For example, nonnegative query charges and shared setup fees give $F(S) = \sum_{q \in S} c_q + \sum_p a_p \mathbf{1}\{S \cap Q_p \neq \varnothing\}$. This illustrates a monotone submodular cost without assuming platform measurements.

2.2 Policies and purchased-set costs

A deterministic adaptive policy purchases an unpurchased query using only purchased answers and stops at a pure feasible state. A *priority policy* fixes a permutation π of Q first. At each position it can TAKE the query, permanently SKIP it, or stop, using only observed answers. Skipping reveals nothing and cannot be undone. The order may depend on the instance and its costs, but not on unobserved answers.

Let $F : 2^Q \to \mathbb{R}_{\geq 0}$ satisfy $F(\varnothing) = 0$ and $F(S) \leq F(T)$ for $S \subseteq T$. For a policy T, let $P_T(w)$ be its final purchased set in world w, and define

$$C_F(T) = \max_{w \in W_A \cup W_B} F(P_T(w)).$$

The optimal adaptive and priority values are A_F and P_F. Policy-class inclusion gives $\mathsf{A}_F \leq \mathsf{P}_F$. For additive prices $\kappa \geq 0$, with $F(S) = \sum_{q \in S} \kappa_q$, write $\mathsf{A}^*(\kappa)$ and $\mathsf{P}^*(\kappa)$.

The main deterministic objective excludes noisy observations, randomization, allowed errors, expected costs, and charges depending on answer values or acquisition order. Appendix I defines a separate zero-error randomized extension with expected purchased-set cost; no randomized conclusion is implicit in the deterministic results. Repeating a physical coordinate is unnecessary with reliable answers. Computational costs of constructing a policy are separate from its query costs.

2.3 Envelopes and dominance

The envelope $E(T)$ is the family of inclusion-maximal sets among $\{P_T(w) : w \in W_A \cup W_B\}$. It is an antichain, with $C_F(T) = \max_{S \in E(T)} F(S)$. For envelopes write $E_1 \preceq E_2$ if every member of E_1 is contained in some member of E_2; equivalently, $\downarrow E_1 \subseteq \downarrow E_2$.

Lemma 1 (Cost-independent dominance) *If $E_1 \preceq E_2$, then E_1 costs no more than E_2 for every monotone F. This relation is necessary as well for domination at every nonnegative additive price vector.*

Proof Containment proves sufficiency. If $S \in E_1$ is contained in no member of E_2, give the coordinates in S unit prices and the others zero prices. The first envelope costs at least $|S|$; the second costs at most $|S| - 1$. □

The recursion combines every query and feasible child-envelope choice, then retains envelopes minimal under $\preceq$. A fixed-order recursion also permits SKIP. Induction on remaining coordinates proves completeness; adjoining a common query and uniting branch families preserve dominance. Appendix A specifies the recurrence, empty-family conventions and paid-context costs.

Lemma 2 (Universal equality criterion) *Equality $\mathsf{A}_F = \mathsf{P}_F$ holds for every admissible monotone F if and only if each minimal adaptive envelope has a realizable priority envelope dominated by it.*

Proof Every adaptive envelope has a minimal adaptive envelope below it, since the family is finite. A priority certificate below that envelope costs no more, proving $\mathsf{P}_F \le \mathsf{A}_F$. Conversely, suppose a minimal adaptive envelope E has no priority envelope below it. Set $F(S) = 0$ on $\downarrow E$ and $F(S) = 1$ elsewhere. This is normalized and monotone. The adaptive witness costs zero, while every priority policy has a path outside $\downarrow E$ and costs one. This converse concerns arbitrary monotone costs, not additive costs alone. □

Incompatible branch orders in one adaptive tree do not rule out a different optimal priority tree.

2.4 Compatible completions

For a labeled promise W, let $\mathrm{Comp}(W)$ contain its compatible total Boolean functions. Let $O_F(g, \pi)$ be the minimum full-domain worst purchased-set cost of a π-ordered decision diagram computing g, and $R_F(g)$ the corresponding minimum over read-once branching programs, without diagram-size restrictions. Then

$$\mathsf{P}_F(W) = \min_{g \in \mathrm{Comp}(W)} \min_{\pi} O_F(g, \pi), \qquad \mathsf{A}_F(W) = \min_{g \in \mathrm{Comp}(W)} R_F(g).$$

Unfolding and completing promise policies proves these identities; Appendix A.1 gives the cost-preserving argument.

The minima may use different completions; answer-equivalent coordinates retain separate cost identities. The appendix addresses paid-context costs, without asserting scalar Bellman optimality.

2.5 Certificate competition and minimax transfer

Fix nonnegative additive prices and write $c(S) = \sum_{q \in S} c_q$, with $F = c$. For $w \in W$, let $C^c_W(w)$ be the minimum cost of a coordinate set whose answers at w determine its label within W, and put $K^c_W = \max_{w \in W} C^c_W(w)$. Every terminating correct policy path is a certificate. Hence

$$K^c_W \le \mathsf{A}_c(W) \le \mathsf{P}_c(W).$$

If a legal priority policy T satisfies $c(P_T(w)) \le \rho C^c_W(w)$ for every $w \in W$, with finite $\rho \ge 1$, then

$$\mathsf{P}_c(W) \le C_c(T) \le \rho K^c_W \le \rho \mathsf{A}_c(W).$$

The multiplicative formulation includes zero certificate costs. A certificate-competitive lower bound alone gives no adaptivity separation: with unit prices, Boolean OR has optimal certificate ratio m, but both minimax values equal m.

For an arbitrary compatible total completion g, every full-domain certificate is a promise certificate, so $C^c_g(w) \ge C^c_W(w)$ on W. The reverse inequality need not hold: for $W = \{00 \mapsto 0, 11 \mapsto 1\}$ with unit prices, both promise certificate costs are one, whereas the completion $g(x) = x_1 \wedge x_2$ has $C^c_g(11) = 2$. One sufficient transfer route is $C^c_g(w) \le \alpha C^c_W(w)$ on W, together with a legal priority implementation of a ρ-certificate-competitive policy for g on the same menu and prices; restriction to W then gives $\mathsf{P}_c(W) \le \rho\alpha \mathsf{A}_c(W)$.

Pointwise certificate preservation is not necessary. Complete an optimal adaptive promise tree as in Appendix A.1, obtaining g_T and a full-domain tree of worst cost $\mathsf{A}_c(W)$. Each of its paths certifies g_T, so $\max_{x\in\{0,1\}^m} C^c_{g_T}(x) \le \mathsf{A}_c(W)$. Thus a legal priority policy for this particular g_T with certificate factor ρ would also give $\mathsf{P}_c(W) \le \rho\mathsf{A}_c(W)$. The remaining issue is whether g_T belongs to a function family with such a uniform guarantee and an admissible global-order implementation. The completion identities do not supply that membership or implementation. Applying the priced-information results to our boundary therefore still requires a menu- and cost-preserving bridge; additive guarantees do not automatically cover shared submodular costs. These transfer observations are supporting comparisons, not claims of new competitive-analysis results.

The two certificate comparisons are quantitatively different on our fixed source. Appendix G proves that the least uniform pointwise inflation over *all* compatible completions is exactly 2 at prices $(2, 2, 3, 0, 1, 1, 1)$, although a suitable completion has maximum full-domain certificate cost 4. It also gives a strict positive-price pointwise obstruction and an exact maximum-certificate bound for read-once AND/OR completions. These are specific transfer limitations, not exclusions of every classical algorithm or reduction.

3 The first possible separation

Theorem 3 (Computer-assisted six/seven boundary) *For complete partition fibres, if* $|Q| \le 6$*, then* $\mathsf{A}_F = \mathsf{P}_F$ *for every normalized nonnegative monotone* F*. A seven-query complete-fibre instance has a strict gap with strictly positive additive prices. Thus seven is the smallest menu size permitting a strict gap for each of the monotone, monotone submodular, nonnegative additive, and strictly positive additive cost classes.*

The order may depend on F; one order optimal for all costs is not asserted. The theorem minimizes queries, not records or classes. We separate the structural reduction, the finite premise, and the explicit upper-size witness.

3.1 A finite catalogue and its structural meaning

Collapse prior SAME blocks on one side. Retain blocks containing query endpoints and their induced DIFFERENT constraints. Every consistent endpoint partition extends to omitted blocks by assigning each a fresh class. Clone shared endpoints and express their identities through prior SAME constraints. At most $2m$ named endpoint positions therefore suffice, without changing physical query coordinates.

After contraction, the menu is a labeled multigraph $G = (V, Q)$, possibly with loops and parallel edges; let D be the forbidden terminal pairs. A set $X \subseteq Q$ of SAME answers is feasible exactly when

$$\mathrm{cl}_{M(G)}(X) = X \quad \text{and no pair in } D \text{ is connected in } (V, X). \tag{1}$$

Transitivity and the negative constraints prove necessity. Assigning each component of (V, X) a separate class proves sufficiency, with omitted blocks assigned fresh classes. This is an equality of answer supports, not a bijection of hidden partitions.

Equivalently, add an auxiliary edge for each forbidden pair and fix its SAME answer to zero. The legal supports are fixed-coordinate sections of the augmented graphic-flat relation. Auxiliary edges are constraints, not purchasable queries. Complementing edge answers gives demand-separated multicuts [9, 10]. All transformations preserve the original purchased sets and F.

Each circuit yields Horn implications from its other edges to the last edge; each forbidden terminal path gives a negative Horn clause. The source class is strictly narrower than arbitrary Horn relations. Indeed, suppose nonconstant coordinates satisfy $x_i = 1 \Rightarrow x_j = 1$. Merge just the feasible endpoint blocks of query i, keeping the other blocks separate. Query j can be one in this atomic world only when it joins the same pair of blocks. Hence the reverse implication also holds. A strict one-way positive implication between nonconstant coordinates is impossible.

For comparison, the four-query Horn supports $A = \{0, 2, 3, 6, 7, 11\}$ and $B = \{4, 5, 14, 15\}$, with prices $(1, 2, 2, 2)$ and integer-bit coordinates, have adaptive cost 5 and priority cost 6 for every one of the 24 orders. The first support has $x_0 \Rightarrow x_1$ without the converse. This exact finite example lies outside the complete-source class for every record count, so it does not refute Theorem 3.

The four-query comparison shows why completeness matters: the six-query equality is specific to graphic-flat sections, rather than to general Horn promises or ordered diagrams. Complete-source and 2-CNF relations are also incomparable when visible coordinates are preserved (Appendix B).

3.2 The checked finite premise

For each endpoint identification, enumerate its unrestricted partition-answer support. Intersect it with every distinct collection of forbidden-pair filters, retaining distinct nonempty supports. Iterative intersection, deduplicated by the entire resulting support, covers all conjunctions. For a fixed endpoint identification and realizable X, the components of (V, X) form the finest partition realizing X. A forbidden-pair filter admits X exactly when its endpoints are in different components. This one partition therefore satisfies all filters that individually admit X, justifying their intersection. Pair disjoint supports, then quotient by *simultaneous* coordinate permutations and label exchange. The sides cannot be relabeled independently, and arbitrary answer-bit flips are not used.

Any disjoint pair of catalogue supports has a common physical realization. Take $2m$ distinct records $r_{i,0}, r_{i,1}$ and queries $q_i = (r_{i,0}, r_{i,1})$. Under each label impose the endpoint identifications and forbidden pairs generating that support. These SAME and DIFFERENT constraints may differ by label; the physical coordinates and their cost identities remain fixed.

A coordinate permutation σ transports cost by $F_\sigma(T) = F(\sigma^{-1}(T))$. This preserves purchased-set costs and the four cost classes in Theorem 3. A representative certificate therefore covers its full orbit. Table 2 states the completed finite premise.

Layer	Count
Six-query endpoint topologies	198,091
Distinct nonempty complete one-sided supports	546,735
Disjoint support-pair orbits	2,860,696
Singleton-side orbits	51,048
Remaining orbits with a shared constant coordinate	61,923
Remaining duplicate/complement gate orbits	129,250
Residual orbits with explicit strategy certificates	2,618,475
Uncovered cases	0

Table 2 Complete six-query classification. The last four case groups are disjoint. Counts are accompanied by membership and complete-certificate checks.

Lemma 4 (Finite certificate premise) *Let $\mathcal{C}_m$ be exactly the family of nonempty supports from endpoint identifications and forbidden-pair conjunctions on m labelled query coordinates. For every disjoint pair in $\mathcal{C}_5$, and every disjoint pair in $\mathcal{C}_6$ with neither a singleton side nor a shared constant coordinate, each minimal adaptive envelope E has a correct priority policy T, compatible with one global permutation, with $E(T) \preceq E$.*

The finite check reconstructs every adaptive frontier and verifies supplied policies for legality, order compatibility and containment. It need not enumerate the whole priority frontier. This premise has no prices; Lemma 2 supplies the all-cost consequence. The orbit audit checks source membership, canonicality, uniqueness, weights and an independent Burnside count. Counts alone are insufficient. Duplicate/complement coordinates receive explicit certificates, not a cost-independent representative substitution.

The two analytic groups require short proofs. If one side is $\{a\}$, take an adaptive tree's query sequence on a. Every opposite-label world differs somewhere on that sequence. Buy it in that fixed order, stopping with the opposite label at the first difference. Each purchased set is a prefix of the set on a, so its worst monotone cost is no larger than that of the original tree. This argument works for arbitrary disjoint supports and any menu size.

If a query has a shared constant answer on both sides, remove its nodes from any policy. Purchased sets shrink and the projected supports remain disjoint. Deletion preserves complete fibres by retaining the same records and constraints and deleting only an observation. On a six-query instance the remaining menu has at most five queries. The independently checked five-query foundation covers $9,439$ supports, $34,601$ pair orbits, and $54,021$ priority-envelope witnesses. Apply it and insert the removed coordinate back as a permanent skip. If primed values denote deletion, then

$$\mathsf{A}_F \le \mathsf{P}_F \le \mathsf{P}'_{F'} = \mathsf{A}'_{F'} \le \mathsf{A}_F, \qquad F'(S) = F(S).$$

Smaller menus are padded with shared constant queries, with $F^+(S) = F(S \cap Q)$, then projected back. Positive dummy prices also suffice when proving the strictly positive additive statement, because they need not be bought.

Together, catalogue completeness, the analytic groups, full adaptive-frontier reconstruction and Lemma 2 establish the lower-size assertion. This is a computer-assisted finite premise, not a proof-assistant-kernel theorem. The executable evidence and all dependency layers remain necessary.

3.3 A complete seven-query source

With coordinates $0, \dots, 6$, use the templates

$$A = (\alpha, 1, \alpha, \beta, \alpha, \gamma, \gamma), \qquad B = (\mu, \mu, 0, \nu, \omega, \omega, \nu).$$

Each side retains all eight assignments to its three free bits; these are combinatorial freedoms, not probabilistic independence. A common vector would force $q_1 = 1$, then $\mu = 1$, $q_0 = 1$ and $\alpha = 1$, contradicting $q_2 = 0$.

For a complete realization, take records $0, \dots, 13$ and $q_i = (2i, 2i + 1)$. Under A, impose the SAME blocks

$$\{0, 4, 8\},\ \{1, 5, 9\},\ \{2, 3\},\ \{10, 12\},\ \{11, 13\},$$

with $6, 7$ unconstrained singleton blocks and no DIFFERENT constraints. Under B, impose

$$\{0, 2\},\ \{1, 3\},\ \{6, 12\},\ \{7, 13\},\ \{8, 10\},\ \{9, 11\},$$

with $4, 5$ singleton blocks and $\mathsf{DIFFERENT}(4, 5)$. Unlisted cross-block merges remain legal. Every satisfying partition has the stated template. Conversely, merge the relevant block pair for each free bit equal to one, and give the remaining blocks distinct fresh classes. Every assignment is realized, so the supports are exact complete-fibre projections.

The frontiers below give, for $t > 0$ and $\varepsilon \geq 0$,

$$\kappa = (2t, 2t, 3t, \varepsilon, t, t, t), \qquad \mathsf{A}^* = \min(5t, 4t + \varepsilon), \quad \mathsf{P}^* = 5t.$$

Taking $t = 1$, $\varepsilon = 1/2$ gives strictly positive prices with values $9/2$ and 5, completing the upper-size assertion of Theorem 3. Section 3.5 gives a direct proof of strict separation: for $0 < \varepsilon < 1$, it establishes $\mathsf{A}^* \leq 4 + \varepsilon < 5 = \mathsf{P}^*$ by incompatible necessary orders across two probe answers. The complete frontiers below additionally determine the exact adaptive value and the all-price ratio.

A readable pair of policies.

Let $N_i(L, R)$ mean buy q_i and follow L on answer zero or R on answer one; leaves A, B output the label. The following saved adaptive witness begins with the inexpensive probe:

$$\begin{aligned} L &= N_0\big(N_1(B, A),\ N_5(N_4(B, A), N_6(B, A))\big), \\ R &= N_4\big(N_5(N_6(A, B), A),\ N_2(B, A)\big), \\ T &= N_3(L, R). \end{aligned}$$

Its envelope is $\{11, 28, 57, 105, 120\}$ and its cost at $(2, 2, 3, 1/2, 1, 1, 1)$ is $9/2$. A priority policy in order $0, 1, 2, 3, 4, 5, 6$ is $N_0(N_1(B, A), N_2(B, A))$, skipping q_1 after $q_0 = 1$. Its envelope $\{3, 5\}$ costs 5. These policies show achievable values and the role of the probe; completeness of the frontiers, not a conflict between these particular trees, proves optimality.

3.4 Compatible formulas and full-domain cost

The fixed source has compatible read-once AND/OR formulas, but this representation need not preserve optimal query cost. List bits in coordinate order $q_0, \ldots, q_6$. The opposite-label pairs

$$(0100000, 1100000), \quad (0100000, 0000000), \quad (1111111, 1101111)$$

have their first member in A and second in B, and differ only at q_0, q_1, q_2, respectively. These three coordinates are therefore essential in every compatible total function. With A positive, monotonicity after answer flips forces q_0 negative and q_1, q_2 positive. Conversely,

$$g(q) = q_2 \vee (q_1 \wedge \neg q_0)$$

is compatible: on A, either $\alpha = 1$ or $(q_1, q_0) = (1, 0)$; on B, $q_2 = 0$ and $q_0 = q_1$. It ignores the last four coordinates. Thus exactly 16 answer orientations admit a monotone completion for this label polarity.

Let $\mathcal{R}(W)$ be the compatible total functions admitting an AND/OR formula in which each physical coordinate appears at most once as a literal; unused coordinates are allowed. This formula restriction is stronger than querying each coordinate at most once per decision-tree path. A read-once AND/OR function is evasive on its essential variables [1]. Directly, answer a queried literal by its parent's neutral value, delete it and suppress unary gates. The residual formula stays nonconstant until its final essential variable is queried. Negated leaves only reverse the returned bit. Hence for nonnegative additive prices, $R_c(g)$ is the sum of all essential-coordinate prices, including zero-price cases. The three necessary coordinates and the displayed formula give

$$\min_{g \in \mathcal{R}(W)} R_c(g) = c_0 + c_1 + c_2.$$

At $(2, 2, 3, 1/2, 1, 1, 1)$ this minimum is 7, whereas the adaptive promise value is $9/2$. An adaptive-optimal completion from Section 2.5 therefore lies outside this formula family at these prices. Other, nonoptimal completions may still support useful promise-restricted certificate bounds. This cost-preservation obstruction concerns the stated family on the original menu; it establishes neither exclusion of every prior reduction nor a practical application advantage. Proposition 12 separately determines the best maximum full-domain certificate cost in this family; Proposition 13 addresses pointwise inflation without a formula restriction.

3.5 A direct order obstruction

We prove directly that the source in Section 3.3, at prices $(2, 2, 3, \varepsilon, 1, 1, 1)$ with $0 < \varepsilon < 1$, satisfies

$$\mathsf{A}^* \le 4 + \varepsilon < 5 = \mathsf{P}^*.$$

The complete-fibre realization and the adaptive and priority upper bounds are already given there. It remains to prove the priority lower bound. This argument does not use the enumerated frontiers and does not replace the six-query classification or the all-price theorem.

Disclose $q_3 = \beta$ for free at the outset, deleting its position but keeping *one common permutation* of all other coordinates for both values of β. An original policy can ignore this information until its former q_3 position and simulate its original decisions without paying for that query. This is therefore a relaxation; earlier pure stopping can only lower cost. Put

$$(x, y, z, \ell, m, r) = (q_0, q_1, q_2, q_4, q_5, q_6), \qquad (c_x, c_y, c_z, c_\ell, c_m, c_r) = (2, 2, 3, 1, 1, 1).$$

The exact conditional supports in this notation are

$$\mathcal{A}_\beta = \{(a, 1, a, a, c, c) : a, c \in \{0, 1\}\}, \qquad \mathcal{B}_\beta = \{(u, u, 0, w, w, \beta) : u, w \in \{0, 1\}\}.$$

Suppose every path costs at most four in this relaxed game. Both labels are initially feasible for either β. Preliminary SKIPs reveal nothing. Opposite-label worlds agreeing on all available coordinates except one force its purchase on their common transcript; if that coordinate was skipped, correctness is impossible. Granting previously skipped coordinates back can only make a lower-bound subproblem easier.

The branch $\beta = 0$ ***forces*** $m \prec \ell$***.***

Use the three legal worlds

$$A_0 = (0, 1, 0, 0, 0, 0), \quad B_0 = (0, 0, 0, 0, 0, 0), \quad B_1 = (1, 1, 0, 0, 0, 0).$$

On A_0, both y and x are necessary to exclude B_0 and B_1, respectively. The following cases exhaust the first actual purchase.

First query	A feasible response defeating budget four
ℓ, m, r	Answer 0. The three displayed worlds remain; the additional x, y cost is four after a unit purchase.
z	Answer 0. The same three worlds require an additional four after paying three.
y	Answer 1. Worlds A_0, B_1 force x, using the entire remaining budget two; any other positive purchase first makes x unaffordable. Yet $x = 1$ still leaves both $(1, 1, 1, 1, 0, 0) \in \mathcal{A}_\beta$ and B_1 feasible, so buying only x cannot finish.

Thus the first purchase must be x. On its feasible answer 1, y is constant and z is unaffordable. The informative unit-coordinate supports, with budget two, are exactly

$$A' = \{100, 111\}, \qquad B' = \{000, 110\} \quad \text{on } (\ell, m, r).$$

No unit coordinate can have been skipped before the first unit purchase: the opposite-label pairs 100/000, 100/110 and 111/110 differ only at ℓ, m, r, respectively. Nor can an endpoint be purchased first. After $\ell = 1$, the surviving worlds are $100, 111$ versus 110; neither remaining coordinate alone separates their labels. After $r = 0$, they are 100 versus $000, 110$, again not separable by either single remaining coordinate. A constant-query purchase only consumes budget. Consequently m must be purchased first among the units, which requires $m \prec \ell$ in the fixed order.

The branch $\beta = 1$ ***forces*** $\ell \prec m$***.***

The legal pair

$$A_+ = (1, 1, 1, 1, 1, 1), \qquad B_+ = (1, 1, 0, 1, 1, 1)$$

differs only at z. Again consider every first purchase other than ℓ.

First query	A feasible response defeating budget four
x, y	Answer 1. The displayed pair forces z, whose price three exceeds the remaining budget two.
z	Answer 0. Budget one remains. Each affordable query has a mixed answer: $\ell = 0$, $m = 0$, or $r = 1$, respectively, as follows directly from the conditional templates. Hence no one-query continuation works.
m, r	Answer 1. The displayed pair forces z, consuming the entire remaining budget three. Any other purchase first makes this common transcript unaffordable. Buying z next also fails: its answer 0 remains mixed, with $A = (0, 1, 0, 0, 1, 1)$ and a matching B (choose $u = 0, w = 1$ after $m = 1$, or $u = 0, w = 0$ after $r = 1$).

Only ℓ can be the first purchase. If $m \prec \ell$, then m must already have been permanently skipped. On the feasible answer $\ell = 0$, take

$$\widehat{A} = (0, 1, 0, 0, 1, 1), \quad \widehat{B}_0 = (0, 0, 0, 0, 0, 1), \quad \widehat{B}_1 = (1, 1, 0, 0, 0, 1).$$

After omitting the unavailable m, these worlds agree outside x, y. On $\widehat{A}$ both heavy queries are necessary, costing four while only three remain. The coordinates z, r supply no substitute; additional prior skips cannot help. Thus m cannot precede ℓ, proving $\ell \prec m$. This step is what rules out using a fixed order with history-dependent permanent SKIPs, rather than merely comparing two particular decision trees.

The two necessary precedences contradict the common remaining permutation. Finally, all six nonprobe prices are integers. An original priority policy of total worst cost strictly below five would induce, after free disclosure, a policy whose every non-probe path costs at most four. The contradiction gives $\mathsf{P}^* \geq 5$. The priority policy in Section 3.3 costs exactly five and ignores q_3; its adaptive policy costs at most $4 + \varepsilon$.

This proves the displayed strict separation without an exact adaptive lower bound or a permutation scan.

4 Sharp additive prices for the fixed witness

Theorem 5 (Fixed-source price bound) *For exactly the source in Section 3.3,*

$$\max_{\kappa \geq 0 : \mathsf{A}^*(\kappa) > 0} \frac{\mathsf{P}^*(\kappa)}{\mathsf{A}^*(\kappa)} = \frac{5}{4}.$$

Equality holds precisely on $\kappa = t(2, 2, 3, 0, 1, 1, 1)$, $t > 0$, in the stated numbering. Over strictly positive prices the supremum remains $5/4$, but is not attained. This is not an upper bound for every seven-query source.

4.1 Complete frontiers

An integer mask encodes a purchased set, with bit q denoting coordinate q. The complete adaptive and priority frontiers are shown in Table 3. Display rows do not pair policies; five envelopes are shared. The producer used feasible-world masks. Its

Adaptive	Priority
$\{3, 5\}$	$\{3, 5\}$
$\{11, 28, 57, 105, 120\}$	$\{11, 28, 121\}$
$\{11, 57, 60, 105\}$	$\{11, 57, 60, 105\}$
$\{19, 20\}$	$\{19, 20\}$
$\{28, 43, 57, 120\}$	—
$\{35, 49, 52\}$	$\{35, 49, 52\}$
$\{67, 84, 113\}$	$\{67, 84, 113\}$

Table 3 Undominated envelopes, with TAKE/SKIP optimization in all $5,040$ orders.

independent checker used $2,187$ ternary partial assignments, saturated copy clauses and frozenset families. It compared every feasibility flag against direct support enumeration, every order's frontier, and all 13 saved witnesses.

Write $\kappa = (a, b, c, z, d, e, f)$ for prices, not source bits. The priority costs are

$$\begin{aligned}
G_0 &= \max(a + b, a + c), \\
G_1 &= z + \max(a + b, c + d, a + d + e + f), \\
G_2 &= z + \max(a + b, a + d + e, c + d + e, a + e + f), \\
G_3 &= \max(a + b + d, c + d), \\
G_4 &= \max(a + b + e, a + d + e, c + d + e), \\
G_5 &= \max(a + b + f, c + d + f, a + d + e + f).
\end{aligned}$$

The two adaptive-only costs are

$$H_1 = z + \max(a+b, c+d, a+d+e, a+e+f, d+e+f),$$
$$H_2 = z + \max(c+d, a+b+e, a+d+e, d+e+f).$$

Thus $\mathsf{P}^* = \min_{0 \le i \le 5} G_i$ and

$$\mathsf{A}^* = \min(G_0, G_2, G_3, G_4, G_5, H_1, H_2).$$

The formulas hold at all real nonnegative prices because the finite frontiers are complete, not because prices have been sampled.

4.2 Algebraic proof

First eliminate the probe price z. For fixed other prices, write $\mathsf{P}^*(z) = \min(U, z+V)$ with $U = \min(G_0, G_3, G_4, G_5)$ and $V = \min(G_1 - z, G_2 - z)$; write $H_i = z + W_i$. If $V \le W_i$, the ratio P^*/H_i is at most one whenever defined. If $V > W_i > 0$, both $U/(z+W_i)$ and $(z+V)/(z+W_i)$ decrease with z whenever their minimum exceeds one. If $W_i = 0$, the path sets cover all six non-probe coordinates, so those prices vanish and $G_0 = 0$. It suffices to bound each P^*/H_i at $z = 0$. This does not assume that H_i remains adaptive-optimal when z changes. A shared adaptive-optimal envelope gives ratio one.

For H_1, normalize $W_1 = 1$. Then

$$a+b,\ c+d,\ a+d+e,\ a+e+f,\ d+e+f \le 1.$$

If G_0, G_2, G_3 all exceed $r > 1$, their potentially large terms imply $a+c > r$, $e > r-1$ and $d > r-1$, respectively. Adding $c+d \le 1$ and $a+d+e \le 1$ yields

$$2 \ge a + c + 2d + e > 4r - 3.$$

At $r = 5/4$ this is impossible. Hence $\mathsf{P}^*/H_1 \le 5/4$.

For H_2, normalize $W_2 = 1$, so

$$c+d,\ a+b+e,\ a+d+e,\ d+e+f \le 1.$$

If G_0, G_4, G_3 all exceed $r > 1$, their terms imply $a+c > r$, $e > r-1$, and $d-e > r-1$. Consequently

$$2 \ge a + c + 2d + e > 6r - 5.$$

Setting $r = 7/6$ is impossible, giving $\mathsf{P}^*/H_2 \le 7/6$ and therefore $\mathsf{P}^*/\mathsf{A}^* \le 5/4$.

At $(a, b, c, z, d, e, f) = (1/2, 1/2, 3/4, 0, 1/4, 1/4, 1/4)$, the formulas give $\mathsf{A}^* = 1$ and every priority envelope costs $5/4$. For global equality, only H_1 can be optimal and strict probe-price decrease forces $z = 0$. Normalize $W_1 = 1$. All priority costs are at least $5/4$, forcing $a + c \ge 5/4$, $d, e \ge 1/4$, and $a + c + 2d + e \le 2$. Equality is necessary throughout. Thus $d = e = 1/4$, $c = 3/4$ and $a = 1/2$. From $G_3 \ge 5/4$

and $a + b \le 1$ obtain $b = 1/2$. The inequality $a + e + f \le 1$ gives $f \le 1/4$, and $G_1 = \max(1, 1, 1 + f) \ge 5/4$ gives the reverse bound. Scaling proves the unique ray.

Every strictly positive vector has $z > 0$. When its ratio exceeds one, removing z strictly increases it, ruling out attainment of $5/4$. The vectors $(2, 2, 3, \varepsilon, 1, 1, 1)$ approach ratio $5/(4 + \varepsilon) \to 5/4$ as $\varepsilon \downarrow 0$. If $\mathsf{A}^* = 0$, all zero-price coordinates together separate the supports and can also be queried in a fixed order at zero cost. Hence $\mathsf{P}^* = 0$ too; such vectors are excluded from the ratio. This completes Theorem 5.

A specified eleven-query extension.

Appendix F gives an explicit complete twenty-two-record realization of an eleven-query extension. Its additive upper bound is $5/4$, inherited by every identifiable coordinate projection; the sixteen menus containing the original seven queries have matching maximum $5/4$ over nonnegative prices. This is a specified-source result. Its separate direct certificate packet covers every price cone, without an old seven/eight-query upper-bound premise.

5 Sharp submodular costs for the fixed witness

Throughout this section the source is exactly the one in Section 3.3. A normalized monotone cost F is submodular if

$$F(S) + F(T) \ge F(S \cup T) + F(S \cap T) \qquad (S, T \subseteq Q).$$

This includes nonnegative query charges with shared setup fees. The complete monotone-cost frontiers in Table 3 remain the finite premise.

Theorem 6 (Fixed-source submodular bound) *For every normalized nonnegative monotone submodular F on this fixed source, $\mathsf{A}_F = 0$ implies $\mathsf{P}_F = 0$. Whenever $\mathsf{P}_F > \mathsf{A}_F$,*

$$3\mathsf{P}_F + F(\{q_3\}) \le 4\mathsf{A}_F. \tag{2}$$

The maximum of $\mathsf{P}_F/\mathsf{A}_F$ over costs with $\mathsf{A}_F > 0$ is $4/3$, already attained by a coverage cost. If F must increase strictly under every proper inclusion, the supremum is $4/3$ and is not attained, both for submodular and for coverage costs.

Proof For this proof only, write $F(0345)$ for $F(\{q_0, q_3, q_4, q_5\})$ and similarly for other coordinate strings; these strings are not decimal masks. Let H_1, H_2 denote the costs of the two adaptive-only envelopes:

$$H_1 = \max\{F(013), F(234), F(0345), F(0356), F(3456)\},$$
$$H_2 = \max\{F(234), F(0135), F(0345), F(3456)\}.$$

Set $Y = 03456$, $X = 2345$, $Z = 014$, $I = 345$ and $J = 34$. Submodularity gives

$$F(0345) + F(3456) \ge F(Y) + F(I),$$
$$F(234) + F(I) \ge F(X) + F(J).$$

For H_1, a third application and monotonicity give

$$F(013) + F(J) \ge F(0134) + F(3) \ge F(Z) + F(3).$$

For H_2, replace this third line by

$$F(0135) + F(J) \geq F(01345) + F(3) \geq F(Z) + F(3).$$

Adding the relevant three lines cancels $F(I)$ and $F(J)$, proving for each $i = 1, 2$ that

$$F(Y) + F(X) + F(Z) + F(3) \leq 4H_i. \tag{3}$$

Suppose $\mathsf{P}_F > H_1$. The priority envelopes with masks $\{11, 28, 121\}$ and $\{11, 57, 60, 105\}$ respectively force $F(Y) \geq \mathsf{P}_F$ and $F(X) \geq \mathsf{P}_F$: all their other path costs are at most H_1, and the cost of any valid priority policy is at least its optimum. The envelope $\{19, 20\}$ forces $F(Z) \geq \mathsf{P}_F$, because $24 \subseteq 234$ implies $F(24) \leq H_1$.

If instead $\mathsf{P}_F > H_2$, the envelope $\{11, 28, 121\}$ again forces $F(Y) \geq \mathsf{P}_F$, since $013 \subseteq 0135$. In $\{35, 49, 52\}$, the paths $015 \subseteq 0135$ and $045 \subseteq 0345$ cost at most H_2. Thus $F(245) \geq \mathsf{P}_F$, and $245 \subseteq X$ gives $F(X) \geq \mathsf{P}_F$. The envelope $\{19, 20\}$ gives the same bound for $F(Z)$. Hence (3) yields $3\mathsf{P}_F + F(3) \leq 4H_i$ whenever $\mathsf{P}_F > H_i$.

If a shared envelope attains A_F, then $\mathsf{P}_F = \mathsf{A}_F$. Otherwise an H_i attains A_F, by completeness of the adaptive frontier. In particular, when $\mathsf{P}_F > \mathsf{A}_F$, selecting such an i proves (2). If $\mathsf{A}_F = 0 < \mathsf{P}_F$, this inequality contradicts nonnegativity. For positive A_F, the ratio is either one or at most $4/3$.

For attainment, take the coverage cost

$$F_0(S) = |S \cap \{q_0, q_1, q_4, q_6\}| + 2\mathbf{1}\{q_2 \in S\} + \mathbf{1}\{S \cap \{q_0, q_5\} \neq \varnothing\}. \tag{4}$$

It has independent query charges and one unit setup charge shared by q_0, q_5; q_3 is free. The seven adaptive envelope costs in Table 3 are $4, 3, 4, 4, 3, 4, 4$, and all six priority envelopes cost four. Thus $\mathsf{A}_{F_0} = 3$ and $\mathsf{P}_{F_0} = 4$.

Strict increase implies $F(\{q_3\}) > 0$ and $\mathsf{A}_F > 0$, since every correct policy makes a nonempty purchase. Equation (2) then rules out ratio $4/3$. Conversely, $F_\delta(S) = F_0(S) + \delta|S|$ is strictly increasing coverage for $\delta > 0$, and the same finite frontiers give

$$\mathsf{P}_{F_\delta} = 4 + 2\delta, \qquad \mathsf{A}_{F_\delta} = \min(4 + 2\delta, 3 + 4\delta).$$

For $0 < \delta \leq 1/2$ the ratio tends to $4/3$ as $\delta \downarrow 0$, proving both suprema. □

6 Unrestricted monotone costs

The finite bounds in Theorems 5 and 6 depend on their respective cost classes. Let $E = \{11, 28, 57, 105, 120\}$ and, for $\varepsilon > 0$, define

$$F_\varepsilon(S) = \varepsilon|S| + \mathbf{1}\{S \notin {\downarrow}E\}.$$

It is normalized and strictly increasing under proper inclusion. The six priority envelopes have costs

$$1 + 2\varepsilon,\ 1 + 5\varepsilon,\ 1 + 4\varepsilon,\ 1 + 3\varepsilon,\ 1 + 3\varepsilon,\ 1 + 4\varepsilon.$$

Envelope E costs 4ε, and the other adaptive-only envelope costs $1 + 4\varepsilon$. Therefore

$$\mathsf{P}_{F_\varepsilon} = 1 + 2\varepsilon, \qquad \mathsf{A}_{F_\varepsilon} = \min(4\varepsilon, 1 + 2\varepsilon).$$

For $0 < \varepsilon \leq 1/2$ the ratio is $1/(4\varepsilon) + 1/2$, unbounded as $\varepsilon \downarrow 0$, with positive denominators. The finite inside/outside cardinality profiles prove these formulas for all real $\varepsilon \geq 0$; rational software controls are not a sampling argument.

These costs are not submodular. Masks $3, 4$ and their intersection 0 belong to $\downarrow E$, but their union 7 does not. The modular term cancels and the submodular inequality fails by one. The construction thus establishes no unbounded additive or submodular ratio. It applies the standard downset separation in Lemma 2, rather than introducing a new general lower-bound technique.

7 Scope and reproducibility

The boundary concerns menu size; ratio bounds are restricted to the specified seven-query source and eleven-query extension. We assert no polynomial-time synthesis or practical acquisition advantage. Academic-evidence use would require validated executable queries, a defensible complete promise, and same-information baselines including reads, joins, batching and caching. We claim no gain in rankings, semantic judgement or historical authentication.

The acquisition unit matters even on this fixed source. Appendix E replaces paid pair answers by paid record reads that reveal every equivalence relation among acquired records. Splitting each price in $(2, 2, 3, 1/2, 1, 1, 1)$ equally between its two endpoints preserves the cost of acquiring that isolated pair, but the new adaptive and priority optima are both $15/4$. Thus the pair-menu gap does not survive this particular change of information and cost contract. This finite control supplies no measured file-system or scientific-workflow benefit. For a separate even-length alternating source family, Theorem 8 gives a priority normal form for every monotone purchased-set cost. The appendix also gives an exact additive-cost construction. These supporting model comparisons use classical path-search ideas and do not strengthen the six/seven pair-query boundary.

Appendix H reports a prospective Wine execution and an offline audit of deposited VPS3 artifacts. These cases distinguish observed execution, numerical compatibility and historical provenance. A same-information fixed verifier also accepts the Wine receipt, and ordinary fixed retrieval and arithmetic suffice for VPS3. Neither case establishes a complete-fibre promise for a natural task or an adaptive acquisition advantage.

Appendix D separates the complete six-query computational layers and their analytic premises from the fixed-seven supplement. The latter rebuilds $5,040$ order frontiers and checks 13 trees plus both 864-certificate families; coverage checks supplement the general submodular proof. The accompanying versioned ledger records the completed original V03 aggregate run and the narrower acceptance of the later frozen candidate.

Statements and Declarations

Author information and contributions.

Names, author order, affiliations, contact information, contribution statements and final responsibility assignments have not been supplied or approved for this draft. The title-page text is a preparation placeholder, not a proposed author identity.

Funding and competing interests.

Author declarations are pending. No statement of absence of funding or competing interests is made.

Research process and generative AI.

Generative AI tools, including Codex agents, were used in research exploration, proof development, implementation of search and verification programs, automated cross-checking, manuscript drafting and formatting. Programmatic certificate replay and agent cross-checks are not human peer review. Technical checking is performed with the stated programmatic and AI-assisted methods. Actual author information, final disclosures and informed approval by the responsible human authors remain pending. Before submission those authors must understand the final claims, approve the manuscript and accept responsibility for it; this statement does not impose a separate manual replay of every technical check. Combinatorial cases are not independent scientific studies or human gold labels.

Data and code availability.

The complete boundary evidence and fixed-seven supplement are available in the public review archive. Appendix D identifies the fixed delivery record, version correspondence and verification scope. The separate Wine and VPS3 packages are public through the versioned records in Appendix H; executed source commits and later archive commits are distinguished. The fourteen-record control has a separate public record-acquisition supplement. The full-eleven direct-proof packet has a separate public source record and a completed Linux replay, with retained hosted receipts; these are separate from the frozen original review archive. The certificate-transfer supplement has a separate public source packet and retained Linux replay receipts. They are not part of the frozen original review archive; Appendix G specifies their mathematical and execution scope. Appendix F states its source, direct certificate obligations and execution scope. The optional even-path calibration and additive-solver implementation are prepared separately; they are not included in the frozen review archive. The even-path theorem has an analytic proof independent of this calibration. The randomized small-menu extension in Appendix I has a separate private proof-object and standard-library replay packet, bound by the current manuscript evidence ledger. It is not included in these earlier public archives. Public access does not grant a blanket project licence. The responsible authors must finalize applicable distribution terms and author declarations before submission. Source-specific notices accompany the materials; a deposited dataset's licence is not a blanket repository licence.

Other declarations.

Human-participant, consent and other ethical declarations must be assessed and completed by the responsible authors as applicable; this preparation text does not certify those determinations.

References

[1] Charikar, M., Fagin, R., Guruswami, V., Kleinberg, J., Raghavan, P., Sahai, A.: Query Strategies for Priced Information. Journal of Computer and System Sciences **64**(4), 785–819 (2002) https://doi.org/10.1006/jcss.2002.1828

[2] Cicalese, F., Laber, E., Saettler, A.M.: Diagnosis determination: decision trees optimizing simultaneously worst and expected testing cost. In: Proceedings of the 31st International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 32, pp. 414–422. PMLR (2014). https://proceedings.mlr.press/v32/cicalese14.html

[3] Ebendt, R., Drechsler, R.: On the Sensitivity of BDDs with Respect to Path-Related Objective Functions. In: 2006 IEEE International Symposium on Circuits and Systems. IEEE (2006). https://doi.org/10.1109/ISCAS.2006.1693784 . https://agra.informatik.uni-bremen.de/doc/konf/on-the-sensitivity-iscas06.pdf

[4] Ebendt, R., Drechsler, R.: Exact BDD Minimization for Path-Related Objective Functions. In: Reis, R., Osseiran, A., Pfleiderer, H.-J. (eds.) VLSI-SoC: From Systems to Silicon. IFIP International Federation for Information Processing, vol. 240, pp. 299–315. Springer (2007). https://doi.org/10.1007/978-0-387-73661-7_19 . https://dl.ifip.org/db/conf/vlsi/vlsisoc2005/EbendtD05.pdf

[5] Oliveira, A.L., Carloni, L.P., Villa, T., Sangiovanni-Vincentelli, A.: An Implicit Formulation for Exact BDD Minimization of Incompletely Specified Functions. In: Reis, R., Claesen, L. (eds.) VLSI: Integrated Systems on Silicon, pp. 315–326. Springer (1997). https://doi.org/10.1007/978-0-387-35311-1_26 . Inspected author manuscript dated 5 September 1996 has the shorter title An implicit formulation for exact BDD minimization. https://sld.cs.columbia.edu/pubs/oliveira_vlsi97.pdf

[6] Sasao, T., Matsuura, M.: BDD Representation for Incompletely Specified Multiple-Output Logic Functions and Its Applications to Functional Decomposition. In: Proceedings of the 42nd Design Automation Conference, pp. 373–378 (2005). https://doi.org/10.1109/DAC.2005.193837 . https://www.cs.york.ac.uk/rts/docs/DAC-1964-2005/papers/2005/dac05/pdffiles/p373.pdf

[7] Jiang, Y., Matic, S., Brayton, R.K.: Generalized Cofactoring for Logic Function Evaluation. In: Proceedings of the 40th Annual Design Automation Conference, pp. 155–158. ACM (2003). https://doi.org/10.1145/775832.775873 . https://ptolemy.berkeley.edu/projects/embedded/Alumni/wjiang/proj/dac2003e.pdf

[8] Greiner, R., Hayward, R., Jankowska, M., Molloy, M.: Finding Optimal Satisficing Strategies for And-Or Trees. Artificial Intelligence **170**(1), 19–58 (2006) https://doi.org/10.1016/j.artint.2005.09.002 . Theorem numbering follows the author manuscript dated 9 September 2005

[9] Bérczi, K., Boros, E., Makino, K.: Matroid Horn Functions. Journal of Combinatorial Theory, Series A **203**, 105838 (2024) https://doi.org/10.1016/j.jcta.2023.105838 . Article 105838

[10] Horňáková, A., Lange, J.-H., Andres, B.: Analysis and Optimization of Graph Decompositions by Lifted Multicuts. In: Proceedings of the 34th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 70, pp. 1539–1548. PMLR (2017). https://arxiv.org/pdf/1503.03791v6

[11] Lovász, L., Naor, M., Newman, I., Wigderson, A.: Search Problems in the Decision Tree Model. SIAM Journal on Discrete Mathematics **8**(1), 119–132 (1995) https://doi.org/10.1137/S0895480192233867

[12] Liu, X., Mukherjee, S.: Tight Query Complexity Bounds for Learning Graph Partitions. In: Proceedings of Thirty Fifth Conference on Learning Theory. Proceedings of Machine Learning Research, vol. 178, pp. 167–181. PMLR (2022). https://proceedings.mlr.press/v178/liu22a/liu22a.pdf

[13] Black, H., Mazumdar, A., Saha, B.: Learning Partitions with Optimal Query and Round Complexities. In: Proceedings of Thirty Eighth Conference on Learning Theory. Proceedings of Machine Learning Research, vol. 291, pp. 344–374. PMLR (2025). Inspected theorem text: arXiv:2505.05009v1. https://proceedings.mlr.press/v291/black25b.html

[14] Kulkarni, R., Santha, M.: Query Complexity of Matroids. In: Spirakis, P.G., Serna, M. (eds.) Algorithms and Complexity: 8th International Conference, CIAC 2013. Lecture Notes in Computer Science, vol. 7878, pp. 300–311. Springer (2013). https://doi.org/10.1007/978-3-642-38233-8_25 . https://eccc.weizmann.ac.il/report/2012/063/download

[15] Berger, B., Ezra, T., Feldman, M., Fusco, F.: Pandora's Problem with Combinatorial Cost. Mathematics of Operations Research **50**(4), 3161–3189 (2025) https://doi.org/10.1287/moor.2023.0248 . Published online in 2024. Theorem numbering follows arXiv:2303.01078v1

[16] Huffstutler, T., Kapshikar, U., Miloschewsky, D., Podder, S.: A Framework for Ruling Out Quantum Speedups. arXiv:2603.29256v3, revised 30 September 2026 (2026). https://doi.org/10.48550/arXiv.2603.29256 . https://arxiv.org/abs/2603.29256v3

[17] Chattopadhyay, A., Dahiya, Y., Mahajan, M.: Query Complexity of Search Problems. In: 48th International Symposium on Mathematical Foundations of Computer Science (MFCS 2023). Leibniz International Proceedings in Informatics (LIPIcs), vol. 272. Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2023). https://doi.org/10.4230/LIPIcs.MFCS.2023.34 . Article 34, pp. 34:1–34:15. https://doi.org/10.4230/LIPIcs.MFCS.2023.34

[18] Göös, M., Harms, N., Riazanov, A., Sofronova, A., Sokolov, D., Yuan, W.: Pseudodeterministic Communication Complexity. In: Proceedings of the 58th Annual ACM Symposium on Theory of Computing, pp. 2030–2039. ACM (2026). https://doi.org/10.1145/3798129.3800907 . Inspected current text: arXiv:2511.04794v2. https://doi.org/10.1145/3798129.3800907

[19] Watson, T.: Pseudodeterminism and MA $\neq$ $\mathrm{NP}^{\mathrm{BPP}}$ in Communication Complexity. arXiv:2608.26425v1, 26 August 2026 (2026). https://doi.org/10.48550/arXiv.2608.26425 . https://arxiv.org/abs/2608.26425v1

A Explicit policy recurrence

For precision, write $X = (X_A, X_B)$ for the current labeled feasible worlds and U for available coordinates. Let I_q contain the answers with nonempty feasible children. For a selected child envelope E_b at each $b \in I_q$, define

$$\mathsf{Join}_q((E_b)_{b\in I_q}) = \mathrm{Max}_{\subseteq} \bigcup_{b\in I_q} \{S \cup \{q\} : S \in E_b\}.$$

Here $\mathrm{Max}_{\subseteq}$ retains maximal purchased sets inside one envelope, whereas $\mathrm{Min}_{\preceq}$ retains minimal envelopes. At a nonempty mixed state,

$$\mathcal{A}(X, U) = \mathrm{Min}_{\preceq} \bigcup_{q\in U} \{\mathsf{Join}_q((E_b)) : E_b \in \mathcal{A}(X|_{q=b}, U \setminus \{q\})\ (b \in I_q)\}.$$

The pure-state value is $\{\{\varnothing\}\}$; the value of a mixed state with $U = \varnothing$ is the empty family. Thus an impossible policy combination contributes nothing. Empty feasible children are omitted. For a fixed order, at position i take the minimal union of (i) the family from skipping to $i + 1$ at the same X and (ii) these joined feasible-child families for taking $q_{\pi(i)}$ and continuing at $i+1$. These rules also specify zero-price and constant-coordinate cases.

At a common paid context P, future set S is evaluated by $F(P \cup S)$. Equivalently, set $G_P(S) = F(P \cup S) - F(P)$ for $S \subseteq Q \setminus P$. This cost is normalized, nonnegative and monotone, and remains submodular when F is submodular. Subtracting the common constant preserves continuation comparisons for fixed P, but does not remove dependence on the paid context. Downward-closure inclusion is preserved under adjoining a common query and uniting branch families, so dominance pruning remains valid. General F cannot be replaced by an additive remaining-cost recursion. Empty answer children contribute no legal path.

A.1 Compatible-completion proof

To extend a promise policy, unfold its tree and terminate a branch when no promised world remains. At the first such branch, a feasible sibling world purchases a superset of that branch's purchased coordinates. Monotonicity bounds its cost. The completed tree computes a compatible function, and preserves the order or read-once property. Conversely, restrict a diagram to W, unfold it and stop at pure states. Cost cannot increase. These transformations prove both identities.

B Relation to 2-CNF promises

The source class is not contained in 2-CNF relations: a three-edge path with DIFFERENT endpoints allows every assignment except 111. The majority of legal $110, 101, 011$ is illegal. All 2-CNF relations, including existential projections, are majority-closed. Conversely, $x_0 \Rightarrow x_1$ is 2-CNF but violates the atomic condition. These obstructions

concern representations preserving visible query coordinates; encodings with new paid queries require a separate menu and cost analysis.

C Exact rational corroboration

C.1 Independent rational certificates

For the additive costs H_i of Theorem 5, normalize every path cost to at most one. Choose one path per priority envelope and bound a ratio variable by its cost. The 432 combinations per H_i yield 864 linear programs. Their maximal optima equal $\max_{\kappa \geq 0, H_i \leq 1} \mathsf{P}^*$ and, by homogeneity, $\sup_{\kappa \geq 0, H_i > 0} \mathsf{P}^*/H_i$; $H_i = 0$ implies $\mathsf{P}^* = 0$. An individual branch optimum is not the global adaptive ratio.

The original floating-point solver proposed candidates. Acceptance required exact rational primal and nonnegative dual feasibility, exact column equations and equal objectives. The independent checker rebuilt every program and verified all 864 certificates using rational arithmetic without an LP solver. The two maxima were 5/4 and 7/6. These checks support the algebraic proof without enlarging its source scope.

Independent coverage certificates.

A separate bounded check represents coverage costs as $F(S) = \sum_{\varnothing \neq R \subseteq Q} w_R \mathbf{1}\{S \cap R \neq \varnothing\}$, with 127 nonnegative resource weights. The same 432 path choices for each adaptive-only envelope produce 864 LP branches; their maxima are both 4/3. Every branch has an exact rational primal/dual certificate. A separate checker reconstructs all intersection columns and verifies these certificates without importing the producer or using an LP solver. This certifies the coverage subclass. The three submodular inequalities above, not the coverage LP, prove the upper bound for all submodular costs. Both arguments depend on the already established fixed-source frontiers.

D A map of the computational proof

The six-query catalogue splits into 112,971 analytic and 2,747,725 explicit cases. The analytic cases are 51,048 singleton-side pairs and 61,923 remaining shared-constant pairs; the latter depend on the complete five-query foundation. The explicit cases are 129,250 remaining duplicate/complement pairs and 2,618,475 residual pairs. These disjoint scopes must not be replaced by a sample, a count-only test or a selected-policy comparison.

Obligation	Required evidence
Source completeness	Enumerate endpoint identifications and all distinct forbidden-filter intersections; compare the full support family.
Pair completeness	Genuine source membership, canonical representatives, uniqueness, orbit weights and independent Burnside count.
Five-query foundation	Complete supports and pair orbits; rebuild every adaptive frontier and verify all priority witnesses.
Analytic cases	Singleton proof and constant-coordinate deletion, with monotone paid-context costs.
Explicit six-query cases	Reconstruct full adaptive frontiers and replay all increasing-order TAKE/SKIP certificates for $2,747,725$ cases.
Coverage alignment	Match each physical pair index to its assigned case group and certificate; require no omissions or duplicate accepted ranges.
Fixed-source cost bounds	Rebuild the seven-query frontiers; verify additive and coverage certificates; use algebra and submodularity for the general bounds and nonattainment.

The original V03 portable bundle completed one fresh aggregate run on 11 October 2026. It executed the complete source catalogue, pair-orbit census, five-query foundation and all $2,747,725$ explicit six-query certificates in one invocation. Each stage exited successfully and met its fixed result contract. The explicit layer replayed $4,074,031$ priority witnesses. The terminal status is conditional on the written analytic arguments: neither hashing those arguments nor checking the executable layers constitutes a formal derivation of them. The accompanying iteration-272 closure ledger binds the original bundle, stage receipts, inputs and precise scope. It records an actual local process observation, not an operating-system-signed attestation.

The later frozen candidate retains identical scientific data, mathematical kernels and stage programs. Its aggregate contract changes the textual proof bindings to V08 and omits the historical-index binding; its driver changes only the corresponding contract hash. Its acceptance covers byte inventory, a boundary plan and the fixed-seven supplement. It has not completed a fresh aggregate six-query run under its new contract. The completed original V03 run is evidence about the common computational inputs, not fresh execution of the candidate. This journal-preparation source preserves the V08 mathematical statements and proofs while updating front matter, declarations and this reproduction account; it does not retroactively change either artifact's contract.

The review archive preserves all $5,569$ candidate files and adds an outer source notice, version map and execution-evidence index. Original receipt bytes and derived summaries are labelled separately. The fixed delivery record, at commit `57ed5b6`, binds the four public assets by size and SHA256. The release tag points to source commit `a8f60ce`. An anonymous download and extraction checked all $5,636$ archive members; a relocated plan-only invocation exited successfully. These distribution checks do not add a mathematical replay. The outer correspondence includes journal draft V02, whose mathematical blocks were unchanged in V03. Journal V04

added three analytic explanations (common filter witnesses, a common physical menu, and continuation costs). Journal V05 explained certificate-to-minimax transfer and cited priced-information work. Journal V06 added a self-contained comparison with compatible read-once AND/OR formulas. Journal V07 added the separately checked fourteen-record acquisition control. Journal V08 appends an analytic normal form for a different even-path family and an additive construction in Appendix E, with explicit positioning against classical path search. These revisions do not change the original boundary or pair-menu finite certificate premise, or the inner executable contract binding V08.

For the boundary, the entries are `verify_catalogue_v02.py`, `verify_orbits_v01.py`, `verify_five.py` and `verify_six.py`. From the extracted archive, the aggregate entry is `candidate/boundary/verify_bundle.py`, with `--mode boundary-full --seconds 14400` and a fresh output directory outside `candidate/`. This explicit cap overrides the wrapper's 900-second default. The original bundle V03 run took about $11,199.1$ seconds on its recorded environment; four hours is a cap, not a completion guarantee on another machine. The archive reviewer guide gives the complete command and interpreter requirements. Complete acceptance requires every layer to finish; files-only, planning, smoke or partial results do not suffice. Strategy and rational LP checks use exact integer/rational arithmetic. The orbit census additionally uses NumPy for exact combinatorial operations. A proposed floating-point LP solution is insufficient. Proof inputs and failed executions are preserved. Byte hashes identify versions; they do not establish authorship, human review, historical authenticity or an error-free proof kernel.

The fixed-seven supplement has the entry `verify.py --mode all`. In that invocation, `reproduce_seven.py` rebuilds all $5,040$ order frontiers, checks 13 witnesses and verifies 864 additive certificates; `check_coverage.py` then verifies 864 coverage certificates against the same frontier bytes. It needs Python 3.10+ and its standard library, without an LP solver or model API. Coverage enumeration certifies the coverage subclass; the three-inequality argument in the paper supplies the upper bound for general submodular costs. Exact hashes, executable commands, dependency requirements and accepted versions belong in the accompanying versioned ledger. The public fixed-seven record retains the executed source and results of hosted run 38134598264, attempt 1. That run used source commit `d4903be`; the linked persistent archive is the later commit `5ca862c`. It completed the fixed-seven checks just described, not the universal six-query layers or a machine proof of the general submodular argument.

E Changing the acquisition unit

E.1 A fixed fourteen-record control

Retain the complete 14-record realization in Section 3.3, including every legal cross-block merge. Now an action acquires one previously unread record. Its answer identifies its equivalence class relative to all previously acquired records, with one possible fresh class. The first acquisition conveys no class-name or label information. Only this induced partition is observable: absolute token names, content semantics, sizes and timing are excluded. This is a declared oracle contract, not a measurement of actual

file contents or access time. Both policy classes receive the same responses. A priority policy scans one global record order with causal permanent skips and stopping.

Proposition 7 (Finite record-acquisition control) *On this source, give records* $2i, 2i+1$ *the prices* $c_{2i} = c_{2i+1} = \kappa_i/2$*, where* $\kappa = (2, 2, 3, 1/2, 1, 1, 1)$*. The adaptive and strong-priority worst costs are both* $15/4$*. With unit prices on all records they are both* 4*.*

Proof The A prior has seven SAME components and no required inequality, hence $B_7 = 877$ complete partitions, where B_k is the kth Bell number. The B prior has eight SAME components and requires two specified components to remain different, giving $B_8 - B_7 = 3263$. These full partitions, rather than the eight pair-answer templates per label, define the control.

For the matched prices use integer weights

$$w = 4c = (4, 4, 4, 4, 6, 6, 1, 1, 2, 2, 2, 2, 2, 2).$$

Here the weighted conclusion has a finite computational premise. Let h be a partial induced partition and let $H(h, b)$ denote feasibility within additional integer budget b. It is true at a nonempty pure state. At a mixed state it is true exactly when some unread record r with $w_r \le b$ has

$$H(h \cup \{r \mapsto t\}, b - w_r) = \text{true} \quad \text{for every feasible next class } t.$$

If there is no affordable successful action it is false. Every existing observed class and one fresh class are considered. A class assignment is feasible for a label exactly when unioning its observed equalities with that label's SAME constraints violates neither a required DIFFERENT pair nor an observed inequality. Distinct remaining components provide a completion, proving sufficiency of this test. The recursion terminates because each action adds a record.

Exact evaluation gives $H(\varnothing, 14) = \text{false}$. Integer path costs therefore imply an adaptive lower bound of $15/4$. A saved tree attains integer budget 15; all its paths increase in the single order

$$(8, 7, 5, 4, 1, 2, 0, 3, 9, 6, 10, 12, 11, 13).$$

Scanning this order and skipping until the next node of the tree implements the same observations, leaves and costs. This proves the priority upper bound and hence equality. The independent constraint-extension checker verifies all 83 nodes and 53 leaves of this tree, including exhaustive response coverage and label purity, and evaluates the lower recurrence in 3233 states. A separate full-world implementation corroborates the result on all 4140 partitions. No enumeration of 14! orders is needed. These are program-checked finite claims, not proof-assistant-kernel theorems.

For unit prices, first acquire $0, 1$. If equal, acquire $4, 5$; otherwise acquire $2, 3$. The selected pair is equal under A and different under B. The order $(0, 1, 4, 5, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13)$ implements this four-read policy. For the lower bound, answer the first pair DIFFERENT precisely when it is one of $\{4, 5\}, \{4, 1\}, \{4, 9\}, \{5, 0\}, \{5, 8\}$, and SAME otherwise. For every third record there remains a common induced partition under both labels. The checker verifies this extension assertion for all $91 \cdot 12 = 1092$ first-pair/third-record cases. Thus three reads cannot suffice. □

The accompanying record-acquisition supplement contains the source constraints, integer prices, the saved tree and a standard-library verifier. It checks the weighted lower recurrence, all tree branches and their common order, and the unit-price control. This is a separate finite check from the six-query proof and fixed-seven pair-menu

supplement. The source predicates and isolated-pair prices are preserved, but the action set and available information change. The result neither contradicts the pair-menu theorem nor establishes equality for all record prices, all complete sources or any natural academic-evidence task.

E.2 An all-size record-path family

Consider a separate family on $V = \{0, \dots, n-1\}$, where $n = 2k \geq 2$. The fibre Ω_A contains every partition satisfying $2r \sim 2r+1$ for $0 \leq r < k$. The fibre Ω_B contains every partition satisfying $2r+1 \sim 2r+2$ for $0 \leq r < k-1$, and $0 \not\sim n-1$. All other merges are allowed. The fibres are disjoint: their combined equalities contradict the endpoint inequality. This family differs from the preceding 14-record source; its conclusions do not extend that finite control to all prices.

Use the same relative-record oracle: buying an unread record reveals its equivalences to all previously acquired records. Absolute class names are unavailable. Policies are deterministic and zero-error. A priority policy fixes a record permutation and permits causal permanent skips. Write $S_T(w)$ for the purchased set and $\mathsf{A}_F^{\mathrm{rec}}, \mathsf{P}_F^{\mathrm{rec}}$ for the two optimal worst costs under normalized nonnegative monotone $F : 2^V \to \mathbb{R}_{\geq 0}$. These symbols concern this record model. The first acquisition is uninformative but remains in the charged set.

Theorem 8 (Record-path containment) *For every correct adaptive policy T there is a correct priority policy P such that*

$$\forall w \in \Omega_A \cup \Omega_B \quad \exists h \in \Omega_A \cup \Omega_B : \quad S_P(w) \subseteq S_T(h).$$

Consequently $\mathsf{A}_F^{\mathrm{rec}} = \mathsf{P}_F^{\mathrm{rec}}$ for every such F. The order may depend on T and hence on the chosen cost. The comparison world h need not equal w.

Proof For $j < n-1$, let h_j have two classes $\{0, \dots, j\}$ and $\{j+1, \dots, n-1\}$; let h_{n-1} be all equal. These worlds are legal, with label B for even j and A for odd j. Since n is even, labels alternate cyclically. Retain T's actual first record i, which is fixed before informative observations. No endpoint-first restriction is imposed.

Put $v_t = (i+t) \bmod n$ for $0 \leq t \leq n$, treating $v_n = i$ as a ghost, and define

$$b_t = \mathbf{1}\{v_t \not\sim i\}, \qquad \epsilon_t = \mathbf{1}\{t \geq n-i\}, \qquad g_t = b_t \oplus \epsilon_t.$$

Thus $g_0 = 0, g_n = 1$; the ghost costs no second acquisition. On h_j, writing $p = (j-i) \bmod n$ gives $g_t = \mathbf{1}\{t > p\}$. On this at-most-two-class subfamily, the anchor bits determine all acquired equivalences. The resulting transition search is closely related to inconsistent chain 2-CNF clause search [11, Section 4]: with endpoints fixed, a 0-to-1 edge violates $g_t \vee \neg g_{t+1}$. The classical search structure and the minimax recurrence below are not claimed as new general algorithms.

Restrict T to these n hard worlds. Surviving cuts form an interval $[L, U]$. A read at position t splits it exactly when $L < t \leq U$. Otherwise its bit and complete response are constant on the surviving hard worlds: simulate that response and delete the purchase. Relations involving deleted records remain simulable from their inferred anchor bits on this subfamily. Stop once the interval is singleton. A correct original leaf cannot retain a larger interval, since

adjacent cuts have opposite labels. This produces an alphabetic binary tree with n leaves; normalization only deletes purchases from each original hard-world path.

A split at t leaves possible internal positions $L+1,\dots,t-1$ on the left and $t+1,\dots,U$ on the right. These sets are disjoint. Inductively, each nonanchor record occurs at most once in the whole tree, and therefore exactly once among its $n-1$ internal nodes. The anchor followed by internal-node preorder is a permutation. Every tree path increases in this order; off-path nodes can be skipped once the relevant branch is observed.

Run this tree on any legal, possibly multicolour, world, using only equality to the original anchor. Its g sequence need not be monotone. Maintain $g_L = 0, g_{U+1} = 1$. A response $g_t = 1$ selects $[L, t-1]$, and $g_t = 0$ selects $[t, U]$. The invariant persists, so a leaf p gives adjacent bits $g_p = 0, g_{p+1} = 1$. Both boundary records were acquired, or are the anchor/ghost.

For an ordinary physical edge $(j, j+1)$ the biases agree, including an ordinary edge entering the ghost. Thus the raw bits differ: exactly one endpoint equals the anchor, so the endpoints differ. This excludes A when j is even and B when j is odd. At the seam $(n-1, 0)$ the bias changes from zero to one; a 0-to-1 transition forces both raw bits to be zero. The endpoints then equal the anchor and each other, excluding B. These certificates give precisely the hard-world leaf labels, including endpoint anchors.

Every lifted path reaches the same structural leaf as some h_j, and purchases exactly that normalized leaf set. Hence $S_P(w) \subseteq S_T(h_j)$. Monotonicity bounds the worst cost of P by that of T. Minimizing, and using inclusion of priority policies among adaptive policies, proves equality, including zero and nonadditive costs. □

Corollary 9 (Unit prices) *For $F(S) = |S|$, the common optimum is $1 + \lceil \log_2 n \rceil$.*

Proof After the paid uninformative anchor, the normalized binary tree must distinguish n cuts, requiring height at least $\lceil \log_2 n \rceil$. A balanced alphabetic tree has this height and admits the same full-world lift and priority order. □

Corollary 10 (Nonnegative additive prices) *For $F(S) = \sum_{r \in S} c_r$, $c_r \geq 0$, an optimal priority policy can be constructed in $O(n^3)$ exact arithmetic operations using $O(n^2)$ stored values and splits.*

Proof Let $D[a, \ell]$ be the minimum worst additional price for ℓ consecutive cyclic cuts beginning at a, excluding already acquired boundary records. With all record indices modulo n,

$$D[a, 1] = 0,$$
$$D[a, \ell] = \min_{1 \leq s < \ell} \big(c_{a+s} + \max\{D[a, s], D[a+s, \ell - s]\} \big).$$

The root comparison costs c_{a+s} on both branches; either child tree can be chosen independently. This proves the recurrence by induction. Every alphabetic tree has the preceding lift, and each leaf is attained by its hard world. Containment therefore gives the exact optimum

$$\mathsf{A}_F^{\mathrm{rec}} = \mathsf{P}_F^{\mathrm{rec}} = \min_{0 \leq i < n} \big(c_i + D[i, n] \big).$$

One table filled by increasing ℓ serves every anchor: a continuation depends only on its cyclic interval and internal prices. There are n^2 states and $n^2(n-1)/2$ split candidates.

Saved minimizing splits reconstruct the tree in $O(n)$ time; its anchor and preorder give the causal-skip witness. Execution always compares to the original anchor, even when a subinterval starts elsewhere. Strictly shorter child intervals handle zero-price ties without further assumptions. Exact rational arithmetic additionally depends on input bit length. This is minimax node-priced search, not expected weighted-leaf optimization. □

F A complete eleven-query extension

This appendix concerns one specified complete source and its coordinate projections. All prices here are finite, nonnegative and additive. It does not bound every eleven-query complete source or a source of arbitrary size. Append four coordinates to the templates of Section 3.3:

$$T_A = (\alpha, 1, \alpha, \beta, \alpha, \gamma, \gamma, \alpha, \beta, \beta, \gamma),$$
$$T_B = (\mu, \mu, 0, \nu, \omega, \omega, \nu, \nu, \mu, \omega, \mu).$$

Each label retains all eight assignments to its three free bits.

Complete equivalence realization.

Take records $0, \dots, 21$ and the common physical queries $q_i = (2i, 2i+1)$ for $i = 0, \dots, 10$. Under label A, impose the SAME blocks

$$\{0,4,8,14\}, \quad \{1,5,9,15\}, \quad \{2,3\},$$
$$\{6,16,18\}, \quad \{7,17,19\}, \quad \{10,12,20\}, \quad \{11,13,21\},$$

with no DIFFERENT constraints. Under label B, impose

$$\{0,2,16,20\}, \quad \{1,3,17,21\},$$
$$\{6,12,14\}, \quad \{7,13,15\}, \quad \{8,10,18\}, \quad \{9,11,19\},$$

and $\mathsf{DIFFERENT}(4,5)$, leaving $4, 5$ as the remaining singleton blocks. The source contains every partition satisfying these constraints. All unlisted cross-block merges remain legal.

For each free bit, all left endpoints form one SAME block and all right endpoints form another. Their equality fixes every occurrence of that bit. The fixed-one pair is SAME and the fixed-zero pair is DIFFERENT. Thus every satisfying partition has its label's displayed template. Conversely, start with exactly the displayed blocks, merge the two blocks of each free bit set to one, and give all remaining blocks distinct fresh classes. The groups and pin blocks are disjoint, so this realizes every assignment without violating the DIFFERENT constraint. This proves exact complete-fibre projection. The construction also extends every original fourteen-record partition; new records are only attached to the corresponding existing group blocks.

Write A^*_{11} and P^*_{11} for the adaptive and causal-skip global-order optima on this source.

Theorem 11 (Specified eleven-query additive bound) *For all admissible prices κ,*

$$\mathsf{P}^*_{11}(\kappa) \le \frac{5}{4}\mathsf{A}^*_{11}(\kappa).$$

The maximum ratio over prices with $\mathsf{A}^*_{11} > 0$ is $5/4$. Every identifiable coordinate projection inherits this upper bound. For each of the sixteen menus containing the original seven coordinates, the nonnegative-price maximum and strictly-positive-price supremum are $5/4$.

Proof The upper uses a computer-assisted finite premise. Complete adaptive recursion gives $2,193$ minimal envelopes. Fix one envelope E, let $U = \bigcup_{S\in E} S$, and normalize its worst path cost to at most one. Introduce $\rho \ge 0$ and nonnegative prices on U, with

$$\sum_{q\in S} \kappa_q \le 1 \quad (S \in E), \qquad \rho \le \sum_{q\in U} \kappa_q.$$

The union U separates labels, so buying all U is a correct priority policy. At a branch use a correct U-only ordered policy F and create one child for each of its maximal purchased sets T, adding $\rho \le \sum_{q\in T} \kappa_q$. For the actual optimum $\rho = \mathsf{P}^*_U$, at least one such child contains the point, since P^*_U is at most the worst path cost of F. Thus a complete finite branch tree whose every leaf has a rational dual upper bound $5/4$ proves $\mathsf{P}^*_U \le 5/4$ throughout the normalized domain. The full priority optimum is at most this U-only optimum. Selecting an optimal adaptive envelope and scaling gives the claimed full-menu bound. If $\mathsf{A}^*_{11} = 0$, the union of coordinates used by a zero-cost adaptive tree is free and separates labels, so $\mathsf{P}^*_{11} = 0$.

The delivered direct proof contains 293 legal priority policies and $130,683$ exact rational nodes, covering all $2,193$ envelope roots and all required children. Of these, $5,395$ nodes directly replace 166 roots previously justified by old seven/eight-query results. The self-contained replay rebuilds the complete adaptive frontier from ternary group states, replays all policies on all legal worlds and checks every primal/dual equation and branch disjunction. It imports neither the producer's mathematical functions nor an LP solver. Completeness of the priority-policy library is not a premise of this upper proof.

For the matching lower, take

$$\kappa = (2, 2, 3, 0, 1, 1, 1, 10, 10, 10, 10).$$

The complete full-menu adaptive frontier evaluates to 4. Scalar fixed-order recursion on the seven-coordinate projection checks all $5,040$ permutations and obtains minimum 5, allowing causal permanent SKIP at every observed history. The legal policy $N_0(N_1(B, A), N_2(B, A))$ costs 5. Any policy purchasing an added coordinate on a legal world costs at least 10, whereas an original-seven policy already costs at most 5. Erasing unobserved outside skips preserves a seven-coordinate policy. Hence the full priority optimum is exactly 5 and the ratio is $5/4$.

For an identifiable submenu M, assign every omitted coordinate a price $H > \sum_{q\in M} \kappa_q$. Buying all M is a finite-cost priority policy. A reachable outside purchase cannot improve either optimum; policies avoiding such purchases have precisely the projected histories. Within-label duplicate projections do not alter correctness or worst cost. Both optima therefore equal their M-only values, and the full-source upper transfers. For menus containing all seven original coordinates, restricting the displayed witness gives the same 4 versus 5 lower. Replacing the zero price by $\varepsilon > 0$ and taking $\varepsilon \downarrow 0$ gives the positive-price supremum, by continuity of finite minima of finite maxima of linear policy costs. No positive-price attainment or uniqueness claim is needed. □

The packet `full11_direct_proof_v01.zip` uses Python's standard library. Its scalar recursion compares SKIP with a purchase followed by the worst feasible child,

for one precommitted order at a time. The exact certificates concern every nonnegative real price vector, not a sampled price grid. The relocated Windows replay is accompanied by an actual Linux run: GitHub Actions run 38812275187, attempt 1, executed source commit `37b10299` with Python 3.12.3 and exited with code zero. Its parsed mathematical fields agree with the Windows result; only runtime metadata and elapsed time are excluded. The retained receipt is archived at commit `cba5d748`, which is distinct from the executed source. This is a cross-platform replay of the same checker, not a new independent mathematical implementation. Neither programmatic replay nor this finite-source extension establishes literature priority, human author review, or application advantage.

G Certificate costs under compatible completions

This appendix concerns the complete seven-query source of Section 3.3. All its eight assignments on each side are retained. It quantifies two different certificate comparisons used in Section 2.5; it does not change the deterministic policy objective. Write

$$K_c(g) = \max_{x \in \{0,1\}^7} C_g^c(x)$$

for a compatible total function g. Let $\mathcal{R}(W)$ be the read-once AND/OR formula family of Section 3.4, including signed literals and permitting unused coordinates.

Proposition 12 (Maximum full-domain certificates) *For every nonnegative additive price vector c,*

$$\min_{g \in \mathcal{R}(W)} K_c(g) = \max\{c_0 + c_1, c_0 + c_2, c_1 + c_2\}.$$

At $c^0 = (2, 2, 3, 0, 1, 1, 1)$, the minimum over unrestricted compatible total functions is instead 4. For $c^\varepsilon = (2, 2, 3, \varepsilon, 1, 1, 1)$, where $0 \le \varepsilon < 1$, that unrestricted minimum is at most $4+\varepsilon < 5$, while the displayed read-once minimum is 5. No exact unrestricted positive-ε minimum is asserted.

Proof The three opposite-label Hamming pairs in Section 3.4 force coordinates $0, 1, 2$ to be essential in every completion. Any nominated set of essential leaves in a reduced read-once AND/OR formula can survive a restriction of all other leaves. At a gate whose two children contain nominated leaves, recurse in both. If only one does, set the other child to the gate's neutral value and suppress the gate. A nonconstant child can be set to either value by setting all its signed literals alike. Read-onceness makes these assignments consistent, and every nominated leaf remains essential.

Maximum certificate cost cannot increase under restriction. A certificate at an extended full input, intersected with the surviving coordinates, certifies the restricted function. Its nonnegative cost is no larger. It therefore suffices to consider three-essential-leaf read-once functions on coordinates $0, 1, 2$.

A pure AND or OR has an input forcing all three literals, so its maximum certificate cost is $c_0 + c_1 + c_2$. A mixed three-leaf formula has maximum certificate cost equal to the largest pair sum. For $a \wedge (b \vee d)$, every input has a certificate using at most two leaves, while the

single-success positive inputs force $\{a, b\}$ and $\{a, d\}$, and the double-failure negative input forces $\{b, d\}$. OR duality and literal flips preserve the statement. These cases exhaust the three-leaf formulas. The compatible function $q_1 \wedge (\neg q_0 \vee q_2)$ attains the largest pair sum on the original full cube, proving the first assertion even with zero prices.

For the unrestricted free-probe minimum, interpret the displayed adaptive tree T of Section 3.3 on every Boolean input. Its full-cube worst purchased cost is 4, so each of its paths is a full certificate of cost at most 4. Conversely, the promised input 0100000 has opposite promised Hamming neighbours at coordinates 0 and 1. Every full certificate there buys both, at cost 4. For c^{ε}, the same tree buys coordinate 3 exactly once on each path and gives the upper bound $4 + \varepsilon$. □

For the remainder put $c = c^0$, and abbreviate C_W^c and C_g^c by C_W and C_g. Every promised certificate cost is positive. Define

$$\alpha(g) = \max_{w \in W} \frac{C_g(w)}{C_W(w)}.$$

This maximizes ratios at promised inputs, rather than maximizing full-cube certificate costs before comparison. Its minimizer need not minimize K_c.

Proposition 13 (Sharp pointwise certificate inflation) *Over all compatible total Boolean functions, without a read-once restriction,*

$$\min_{g \in \mathrm{Comp}(W)} \alpha(g) = 2.$$

At prices c^{ε}, for $0 \le \varepsilon < 1$, every compatible total function has pointwise inflation at least $2/(1 + \varepsilon) > 1$. The latter bound is not claimed sharp for positive ε.

Proof Use the promised inputs, in coordinate order,

$$a = 1110111 \in W_A, \quad b = 0000000 \in W_B, \quad d = 1110100 \in W_A.$$

Their promise certificate costs are $1, 2, 2$. At a, the only promise certificate of cost below 2 is $\{3, 6\}$, fixing $(q_3, q_6) = (0, 1)$. Any cheaper single unit-price coordinate, even with the free probe, still agrees with some B input.

Suppose $C_g(a) < 2$, $C_g(b) < 4$ and $C_g(d) < 4$. The certificate at a forces $g = 1$ throughout the full-cube cylinder $q_3 = 0, q_6 = 1$. A certificate at b must contain q_1: the opposite promised input 0100000 differs only there. It must also contain $q_6 = 0$ to avoid that positive cylinder. These two queries cost 3; under a strict budget 4, only free q_3 may be added. The possible coordinate sets are therefore $\{1, 6\}$ and $\{1, 3, 6\}$, fixing $q_1 = q_6 = 0$ and optionally $q_3 = 0$.

The input d agrees at $3, 6$, so its certificate must fix $q_1 = 1$ to avoid this negative cylinder. No promise certificate at d of cost below 4 contains q_1. With that cost-2 query bought, at most one unit-price query among $4, 5, 6$ and the free probe remain affordable; a B input agrees in each case. This contradiction establishes, for every compatible g,

$$C_g(a) \ge 2 \quad \text{or} \quad C_g(b) \ge 4 \quad \text{or} \quad C_g(d) \ge 4.$$

Dividing by the respective promise costs gives $\alpha(g) \ge 2$. This is a three-input cylinder argument over every completion, not an enumeration of all 2^{112} total completions.

For a matching upper bound define the total function

$$g_*(q) = \begin{cases} 1, & q_3 \neq q_6, \\ 0, & q_3 = q_6,\ q_5 \neq q_6, \\ 0, & q_3 = q_5 = q_6,\ q_1 = 0, \\ \neg q_0 \vee q_4, & q_3 = q_5 = q_6 = 0,\ q_1 = 1, \\ \neg q_4 \vee q_2, & q_3 = q_5 = q_6 = 1,\ q_1 = 1. \end{cases}$$

Substituting either source template proves compatibility. Table 4 gives C_W and sufficient full certificates for g_*. Each listed certificate costs at most $2C_W$, so $\alpha(g_*) \leq 2$.

Table 4 Pointwise comparison at c^0. The source parameters are those in Section 3.3; rows with overlapping conditions for B may use either indicated certificate. Coordinate 3 is free. The last column is a certificate-cost upper bound, not a query-policy execution cost.

Label	Parameter condition	C_W	Certificate	Cost
A	$\beta \neq \gamma$	1	$\{3, 6\}$	1
A	$\alpha \neq \beta = \gamma$	2	$\{3, 5, 1, 4\}$	4
A	$\alpha = \beta = \gamma = 0$	4	$\{3, 5, 1, 0\}$	5
A	$\alpha = \beta = \gamma = 1$	3	$\{3, 5, 1, 2\}$	6
B	$\mu = 0$	2	$\{3, 6, 1\}$	3
B	$\nu \neq \omega$	2	$\{3, 5, 6\}$	2
B	$\mu = 1,\ \nu = \omega = 0$	3	$\{3, 6, 0, 4\}$	4
B	$\mu = 1,\ \nu = \omega = 1$	4	$\{3, 6, 4, 2\}$	5

For completeness, the promise-cost entries follow directly from the templates. On A, $\beta \neq \gamma$ is witnessed by $\{3, 6\}$, and $\alpha \neq \gamma$ by $\{4, 5\}$. The free probe alone is never pure. When $\beta = \gamma$, every one unit-price coordinate together with the probe still agrees with some B input, giving the cost-2 lower bound in the second A row. When neither inequality occurs, the all-zero parameter input has Hamming neighbours forcing $0, 1$, and the all-one parameter input has a neighbour forcing 2. On B, $q_1 = 0$ or $q_5 \neq q_6$ yields cost 2. Otherwise the two remaining inputs force, respectively, 0 and one of $2, 4$, or 2 and one of $0, 4$; the certificates $\{0, 4\}$ and $\{2, 4\}$ attain costs $3, 4$. No single unit-price query with the free probe separates these B inputs from all A inputs. This also establishes the cost-2 lower bounds in the preceding cases.

Finally let $0 \leq \varepsilon < 1$. Prices only increase from c^0, so the certificate families affordable below thresholds $2, 4, 4$ in the lower proof only shrink. The same three-way disjunction therefore holds. The promise costs at a, b, d are now $1 + \varepsilon, 2, 2$: at a the unique base certificate below 2 is $\{3, 6\}$, and the other two costs are attained without 3. Dividing the disjunction by these costs proves the asserted lower bound $2/(1 + \varepsilon)$. □

Transfer scope and finite corroboration.

At c^0, a uniform pointwise comparison $C_g(w) \leq \alpha C_W(w)$ requires $\alpha \geq 2$. Composing it with a full-domain certificate-competitive factor $\rho \geq 1$ cannot by itself recover the fixed-source bound $5/4$ through the product $\rho\alpha$. For $0 < \varepsilon < 3/5$, the positive-price lower bound similarly exceeds $5/4$. This does not exclude an input-dependent analysis,

source-aware algorithms, other reductions, or the different maximum-full-certificate bridge with a suitable unrestricted completion and legal priority implementation.

General completion complexity and distinctions between promise-valid and domain-certifying certificates are established context [16]. Our C_W is a priced version of the promise-valid convention. The exact constants above concern this specified complete source; no new general framework or literature priority is asserted.

Separate implementations corroborate the argument by classifying all 64 three-essential-variable read-once truth functions through formula generation and a 256-table cofactor scan, checking 48 affordable certificate triples for opposite-label cylinder conflicts, and checking all 128 outputs of each explicit total witness. The original bounded formula search stopped at a resource limit and is not a proof premise. Independent programmatic replay is neither a proof-assistant kernel nor human author review.

Public replay provenance.

The separately deposited certificate packet was executed once in Linux, Python 3.12.3, by GitHub Actions run 38833749833, attempt 1, using source commit `a420cd488`. The actual checker process exited with code zero. Every parsed mathematical output field agreed with the frozen Windows report; no mathematical field was excluded. The original service-digest-matched artifact and provider metadata are retained at archive commit `f68a04d13`, distinct from the executed source. Attribution to the hosted run is conditional on the stated trust in provider metadata and the artifact digest. This is a cross-system replay of the same checker, not another independent implementation, human author review, or historical scientific authentication.

H Case evidence and applicability limits

These development cases concern research-artifact auditing. They are separate from the complete-source query theorems: neither supplies a complete possible-world model, an independently justified priced physical menu, or a measured adaptive-policy advantage. The counts below are evidence components of selected workflows, not independent scientific studies or human gold labels.

H.1 A selected prospective Wine execution

A fixed author-code configuration was executed in GitHub Actions, run 38136338509, attempt 1, using source commit `6f64dd2`. Across the prescribed 25 seeds, the saved CSV has mean 0.904 and population standard deviation 0.04606811918168868, compatible with the target three-decimal Wine cell. The retained receipt binds the selected execution version, output and hosted-run provenance. Its inspected syscall trace records a successful 10,782-byte read from the designated Wine path, and the retained signature bundle passes the fixed verification policy.

A fixed direct verifier accepts the same artifact. This supports a selected prospective execution under its observation contract; it does not authenticate the historical experiment or establish the complete set of consumed bytes. Earlier failed attempts remain archived. The persistent Wine record at archive commit `ae2ba2d` links the

actual source and retained evidence. The archive commit is not the executed source. No separate baseline training run or policy-cost benefit is claimed.

H.2 An offline VPS3 artifact audit

The selected workflow comes from the Materials Cloud deposit *AMaRaNTA: automated first-principles exchange parameters in 2D magnets [Results]* by Aditya Putatunda, Giuseppe Cuono, Federico Orlando and Silvia Picozzi (DOI: 10.24435/materialscloud:he-3k). The fixed target comprises twelve parameters, 48 calculation jobs and 93 distinct repository objects in one VPS3 workflow. Prespecified raw-input spin states and final sigma-to-zero energies, combined with the declared four-state conventions, reproduce all twelve native parameter values within 10^{-7} meV and agree with the deposited text at its displayed precision. The declared $S = 1$ convention is not an inferred initialization-moment denominator.

All 48 raw energies equal their corresponding stored List-slot finite-decimal values exactly under the statically identified source slot mapping. None of the twelve results agrees with the separate small-JSON central value under the declared comparison. This discrepancy remains undiagnosed; it is not classified as twelve newly discovered physical errors. Since J_{xx} had already been checked from raw files, completion added eleven parameter coverages. Prior retrieval failures and cache reuse remain in the scoped acquisition record, which is not the whole research cost.

The portable verifier re-extracts text and small-JSON values from digest-bound retained source bytes, checks selected typed database ports and statically checks the archived source. It does not execute that author source. The relocated Windows result and the Linux result from hosted run 38140070343, attempt 1, agree apart from their timestamps. The latter uses source commit `3f39c20`; the persistent VPS3 record is at the later archive commit `7ab1b1d`.

This hosted execution runs the offline audit, not VASP or the original workflow. Numerical agreement does not supply the missing CREATE edges, authenticate historical raw-to-List conversion or parser semantics, establish physical convergence, or resolve the small-JSON discrepancy. The package preserves source attribution, the deposit's stated CC BY 4.0 terms and its original public database metadata; it is not an anonymized archive. Ordinary fixed retrieval, relational lookup and arithmetic suffice for these observations. They do not verify the complete-fibre hypothesis for academic evidence or demonstrate practical gains from the query-policy theorems.

I A sharp randomized small-menu boundary

This appendix extends the policy objective explicitly. It does not change the deterministic six/seven comparison. Here labels are written $0, 1$, with complete-fibre projections W_0, W_1 and $W = W_0 \cup W_1$. All physical query identities and the unrestricted number of equivalence classes are retained. Costs $F : 2^Q \to \mathbb{R}_{\geq 0}$ are finite-valued, normalized and monotone.

I.1 Private randomness and a common completion

For $\mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}$, let $\mathsf{D}_{\mathcal{C}}(W; F)$ denote optimal deterministic worst-input cost. Thus $\mathsf{D}_{\mathcal{A}} = \mathsf{A}_F$ and $\mathsf{D}_{\mathcal{P}} = \mathsf{P}_F$. A zero-error randomized policy is an input-independent distribution over correct deterministic policies, with value

$$\mathsf{R}_{\mathcal{C}}(W; F) = \min_{\mu} \max_{w \in W} \mathbb{E}_{T \sim \mu} F(P_T(w)).$$

For priority policies the private seed may sample the permutation before execution; each supported deterministic policy still has a single global order and causal permanent skips. The input is independent of the private seed. We do not use the cost of an expected purchased set, an average over inputs, or a bounded-error criterion.

A set $S \subseteq Q$ is a promise certificate at w if every promised input agreeing with w on S has its label. Put

$$K_F(W) = \max_{w \in W} \min_{S \text{ a promise certificate at } w} F(S).$$

Every correct terminating path is a certificate. Consequently, for both classes,

$$K_F(W) \le \mathsf{R}_{\mathcal{C}}(W; F) \le \mathsf{D}_{\mathcal{C}}(W; F). \tag{5}$$

For a fixed $g \in \mathrm{Comp}(W)$, let $\mathsf{R}^{\mathrm{full}}_{\mathcal{C}}(g; F)$ require correctness and charge worst-input expected cost on the full Boolean cube. Let $\mathsf{R}^{W}_{\mathcal{C}}(g; F)$ retain full-cube correctness but charge only promised inputs. In both definitions all supported trees compute the same g. Then

$$\mathsf{R}_{\mathcal{C}}(W; F) \le \min_{g \in \mathrm{Comp}(W)} \mathsf{R}^{W}_{\mathcal{C}}(g; F) \le \min_{g \in \mathrm{Comp}(W)} \mathsf{R}^{\mathrm{full}}_{\mathcal{C}}(g; F). \tag{6}$$

The cube includes every indexed answer string, even if some strings are not physically realizable for a particular menu. A compatible completion may depend on the source and on F; no single completion is asserted to optimize all costs at once.

I.2 Certificate ideals remove the price quantifier

Theorem 14 (Three-query certificate optimality, computer-assisted) *For every complete equivalence source on at most three physical queries and every finite-valued normalized nonnegative monotone F,*

$$\begin{aligned} K_F(W) &= \mathsf{D}_{\mathcal{P}}(W; F) = \mathsf{D}_{\mathcal{A}}(W; F) \\ &= \mathsf{R}_{\mathcal{P}}(W; F) = \mathsf{R}_{\mathcal{A}}(W; F). \end{aligned}$$

For each source and each such F, one compatible total function and one deterministic priority tree attain $K_F(W)$ on the entire cube. Thus both completion minima in (6) also equal $K_F(W)$, for either policy class. Zero costs are included.

Proof The finite premise is a certificate-ideal cover, which we specify and reduce to complete checked objects below. For every complete source on $m \le 3$ queries and every nonempty downset $I \subseteq 2^Q$, suppose every promised input has a certificate in I. The premise asserts

that there is a total priority tree correct on the promise whose purchased set lies in I on *every full-cube input.*

Complete source reduction. Duplicate shared query endpoints and impose their identities as SAME constraints in both labels. Collapse all original SAME blocks. Endpoint blocks retain every DIFFERENT edge between them. Any endpoint partition satisfying these induced constraints extends to all original records: assign every remaining block without endpoints its own fresh equivalence class. All remaining DIFFERENT constraints are then satisfied. This uses an unrestricted class count, not transitivity of DIFFERENT. Thus arbitrary hidden records impose no extra restriction on endpoint responses.

For three indexed queries this reduces to six endpoints. Their $B_6 = 203$ SAME partitions, each with an arbitrary DIFFERENT graph on its blocks, give

$$\sum_{k=1}^{6} \left\{ {6 \atop k} \right\} 2^{\binom{k}{2}} = 53{,}071$$

satisfiable specifications. Exact projection gives 55 distinct nonempty supports and 924 ordered disjoint two-label source pairs. Query coordinates are never identified or repriced by this reduction.

Exhaustive ideal cover. For dimensions one and two the check covers all partial Boolean tasks with both labels, a superset of the required sources. At dimension three it covers the complete-source catalogue just described. Enumerating every nonempty downset gives the entire case table in Table 5. Each case supplies either an input with no certificate in I, or a global permutation and an explicit total tree. The producer uses quotient-partition enumeration and order recursion. A separate checker reconstructs all 203 closed equivalence graphs from the 2^{15} undirected edge masks, intersects allowed-world bitsets for all 53,071 specifications, and recovers exactly the same 55 supports and 924 source pairs. It independently enumerates downsets and tests certificate availability.

For every applicable case the checker traces the supplied tree on every full-cube input, checks labels on the promise, checks membership of its purchased set in I, and verifies the same global order along all paths. For an inapplicable case it checks the supplied input against all opposite-label agreements to exclude every allowed certificate. All 17,810 source/ideal pairs are covered exactly once. The 10,974 applicable trees pass 87,308 full-cube path checks. Constant tasks, if allowed, use a zero-purchase leaf. No optimizer or floating-point certificate is used in this finite premise.

Queries	Tasks	Nonempty downsets	Cases	Total trees
1	2	2	4	2
2	50	5	250	118
3	924	19	17,556	10,854

Table 5 Complete finite certificate-ideal cover. Dimensions one and two include all partial tasks with two nonempty labels; dimension three includes all complete equivalence sources after projection. These are exhaustive combinatorial cases, not independently sampled studies.

All-cost deduction. Set $t = K_F(W)$ and $I = \{S \subseteq Q : F(S) \leq t\}$. Normalization, nonnegativity and monotonicity make I a nonempty downset. Every promised input has a certificate in I by the definition of K_F. The finite cover supplies a deterministic priority tree

whose every cube path costs at most t. Its leaf function is a compatible total completion. This gives upper bounds t for all stated quantities, while (5) and (6) give the lower bounds. The quantifiers are $\forall W\,\forall F\,\exists(g,T)$, not $\exists g\,\forall F$. The universal price statement follows from this threshold argument, not from extrapolating a finite price grid. □

I.3 Four positive-priced physical queries attain the gap

Corollary 15 (Sharp four-query minimum) *For complete equivalence sources, four is the smallest menu allowing either* $\mathsf{R}_{\mathcal{C}} < \mathsf{D}_{\mathcal{C}}$ *or a strict randomized promise/common-completion gap, for each* $\mathcal{C} \in \{\mathcal{A}, \mathcal{P}\}$. *This holds with strictly positive additive prices, and in the containing normalized monotone cost classes, including monotone submodular costs. Both full-cube and promise-charged common-completion minima have the same minimum menu size.*

Proof Theorem 14 excludes the discrepancies at size at most three. We give the existing four-query witness and its exact analytical values to prove attainability.

Coordinates are listed in order $0, 1, 2, 3$, with positive prices $c = (2, 2, 3, 1)$ and supports

$$W_1 = \{A_0 = 0100, A_1 = 1111\},$$
$$W_0 = \{B_{00} = 0000, B_{01} = 0001, B_{10} = 1100, B_{11} = 1101\}.$$

Use records $r_0, \ldots, r_7$ and disjoint pair query $q_i = [r_{2i} \sim r_{2i+1}]$. For label one the SAME blocks are $\{0, 4, 6\}$, $\{1, 5, 7\}$ and $\{2, 3\}$, with no DIFFERENT constraints. The exact answer form is $(a, 1, a, a)$. For label zero impose SAME blocks $\{0, 2\}$ and $\{1, 3\}$ and DIFFERENT$(4, 5)$, giving exactly $(u, u, 0, w)$. All indicated free bits are realizable. Without the label promise all sixteen strings are physically realizable.

Write $N_i(L, R)$ for query i with zero/one children L, R. Mix the following promise-correct priority trees with masses $1/2, 1/4, 1/4$. Each is total on the cube, but their completions need not agree outside W:

$$U_1 = N_0(N_1(0, 1), N_2(0, 1)),$$
$$U_2 = N_3(N_0(N_1(0, 1), 0), N_2(0, 1)),$$
$$U_3 = N_3(N_1(0, N_0(1, 0)), N_2(0, 1)).$$

Their global orders are $(0, 1, 2, 3)$, $(3, 2, 0, 1)$ and $(3, 2, 1, 0)$ respectively. The expected costs on $(A_0, A_1, B_{00}, B_{01}, B_{10}, B_{11})$ are $(9/2, 9/2, 4, 4, 9/2, 9/2)$. Thus $\mathsf{R}_{\mathcal{P}} \le 9/2$.

For the lower bound assign masses $1/2, 1/6, 1/3$ to A_0, A_1, B_{11}. Opposite-label single-bit neighbours force mandatory query sets $M_{A_0} = \{0, 1\}$ and $M_{A_1} = M_{B_{11}} = \{2\}$. Including the first actual query i in each mandatory set gives average-cost lower bounds $9/2, 9/2, 5, 9/2$ for the four choices of i. That first query is independent of the unknown input. Every correct randomized mixture therefore has worst expected cost at least $9/2$, even in the adaptive class. Deterministic costs are integral, and U_1 attains worst cost 5. Hence, for both classes,

$$\mathsf{R}_{\mathcal{C}}(W; c) = 9/2, \qquad \mathsf{D}_{\mathcal{C}}(W; c) = 5.$$

Now fix a compatible total g and let $z = 0101$. If $g(z) = 0$, opposite-label single-bit neighbours of A_0 force coordinates $0, 1, 3$, costing 5. If $g(z) = 1$, neighbours z, A_1 of B_{11} force coordinates $0, 2$, again costing 5. The input incurring this cost is promised. Every supported deterministic tree computing this fixed g must pay at least 5 on that same input, so randomizing trees cannot remove the lower bound. Finally U_1 computes the compatible total function equal to x_1 when $x_0 = 0$ and to x_2 when $x_0 = 1$, at full-cube worst cost 5. Both common-completion minima therefore equal 5, proving all claimed four-query gaps. □

I.4 Proof provenance, literature and limits

The new three-query finite premise has separately generated proof objects and an independent standard-library checker. Its portable packet has also been extracted and actually replayed at a different local path. This is programmatic, computer-assisted verification, not a proof-assistant kernel or human review. The four-query witness is inherited from the earlier immutable companion result, with its separate certificate report; its sixty closed files were hash-checked when accepting the new small-menu boundary. The older four-query computation was not re-executed for this manuscript revision. The proof above is explicit so that its source, mixture, lower bound and completion collision can be checked directly. The manuscript evidence ledger distinguishes new three-query execution from inherited four-query evidence.

The earlier exploratory scan used 125 partial-task symmetry classes and 64 price triples. Its failure to find a gap is not a proof premise. Theorem 14 instead uses the complete source/ideal cover and the threshold deduction for every monotone F. The three-query theorem is not asserted for every arbitrary partial three-bit function.

Consistent outputs and compatible total completions are established context in pseudodeterministic search and communication [17–19]. The inspected current STOC paper includes a bounded-error query warmup, and Watson gives a zero-sided-error communication separation. These depth/abstention formulations differ from the finite expected purchased-set objective used here. The bounded primary-source comparison does not establish bibliographic priority or rule out a subsuming theorem. The present candidate contribution is the exact minimum physical menu within complete equivalence fibres and the all-monotone certificate identity below that minimum.

The four-query minimum concerns randomization benefit and a common-completion discrepancy, not the deterministic adaptive/order separation. The latter remains six versus seven. Neither the largest four-query ratio nor the first gap between randomized adaptive and randomized priority classes is determined here. No practical academic-evaluation gain, new model experiment, human gold label, or new historical execution certification follows from this appendix.